

Comparing and analysing systems using shape metrics

Sept 15, 2026

Bruna Lopes



Overview

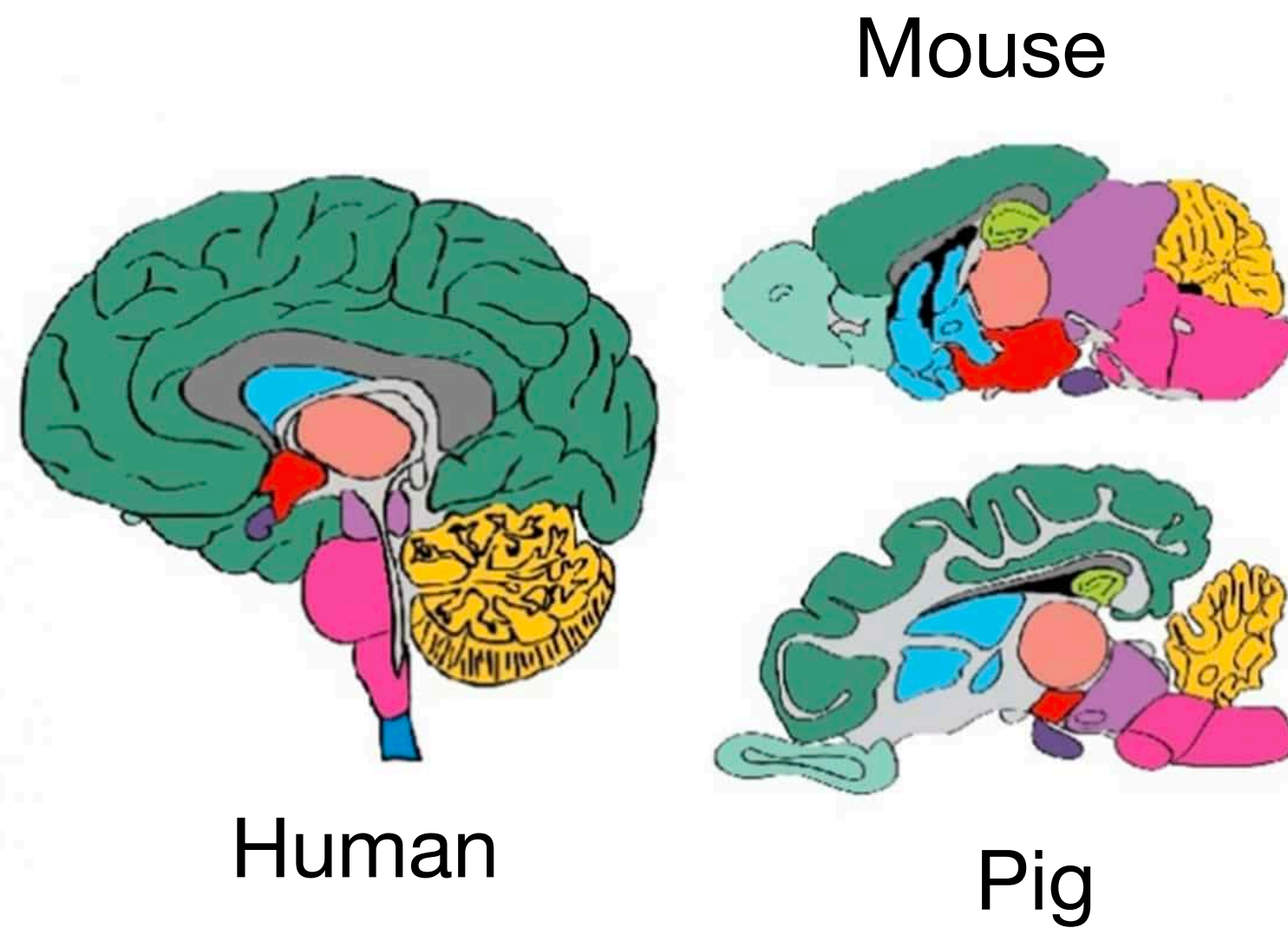
1. **Introduction:** Why do we care about comparisons?
2. **Where to start?** Representing systems
3. **How to?** Shape metrics
 1. Basics of distance functions
 2. Procrustes
4. **Hands on:** Notebook tutorial
 1. Load data
 2. Build representation matrices
 3. Compute distances
 4. Plot representations

How to **understand complex systems?**

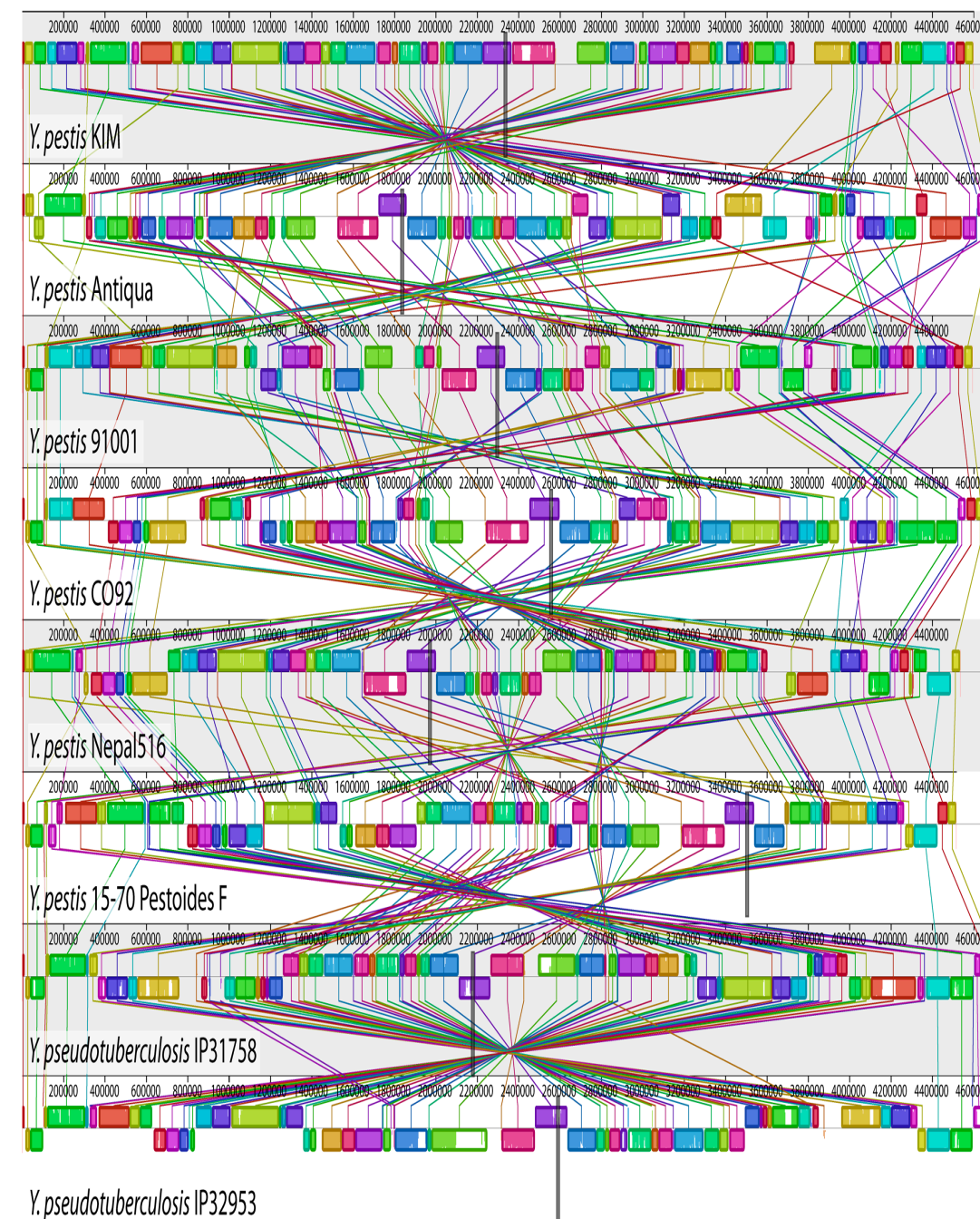
How to understand complex systems?

Through comparisons

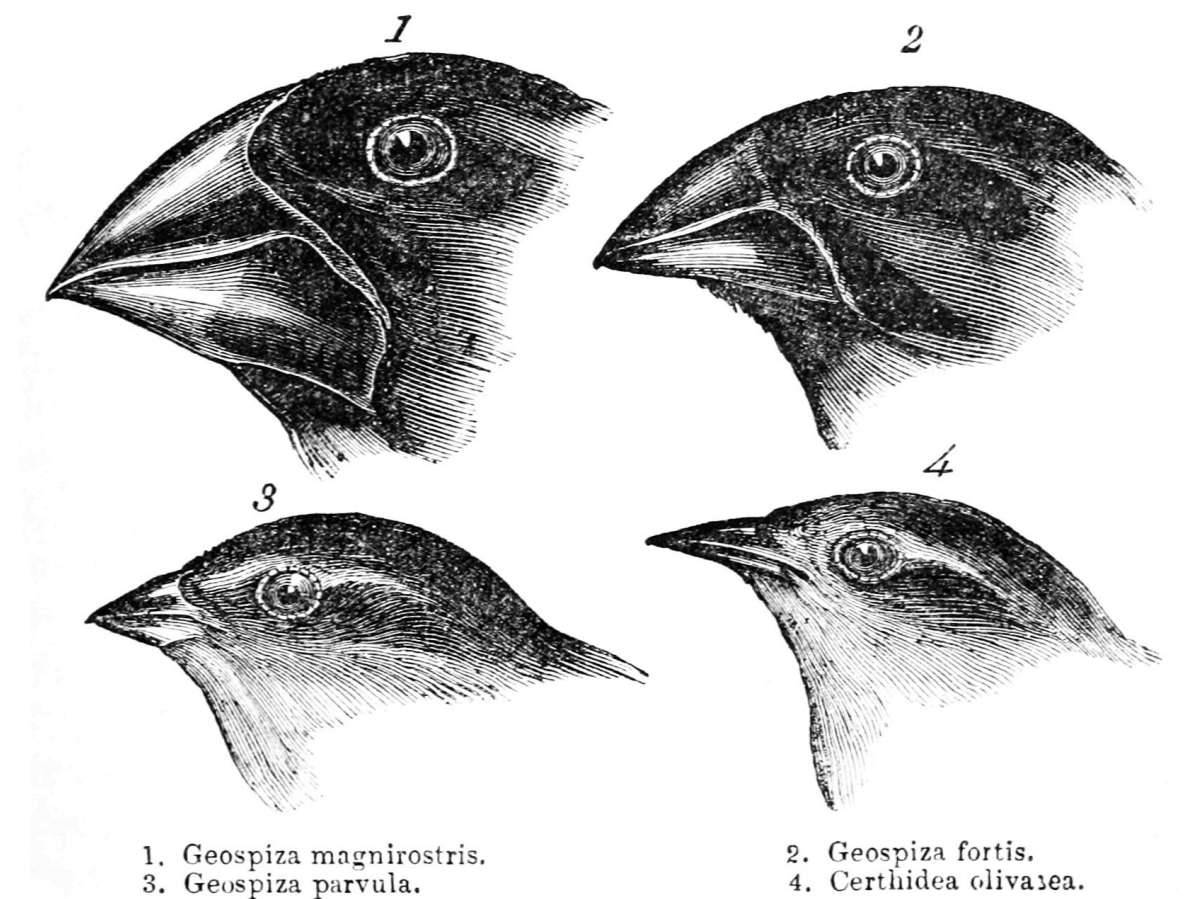
Comparative Anatomy



Comparative Genomics



Comparative Ecology

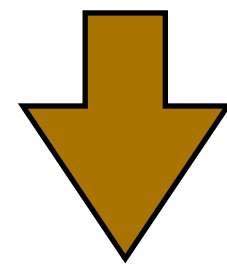


Darwin (1845)

Imagine the situation

Main question:

How similar/different they operate?

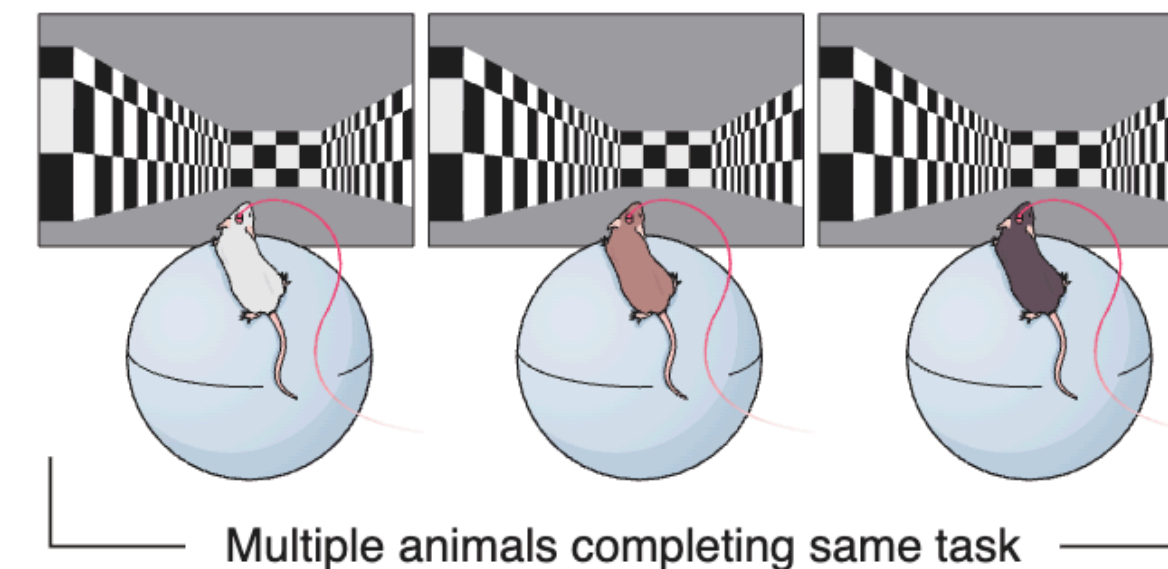


How do they react under the same circumstances?

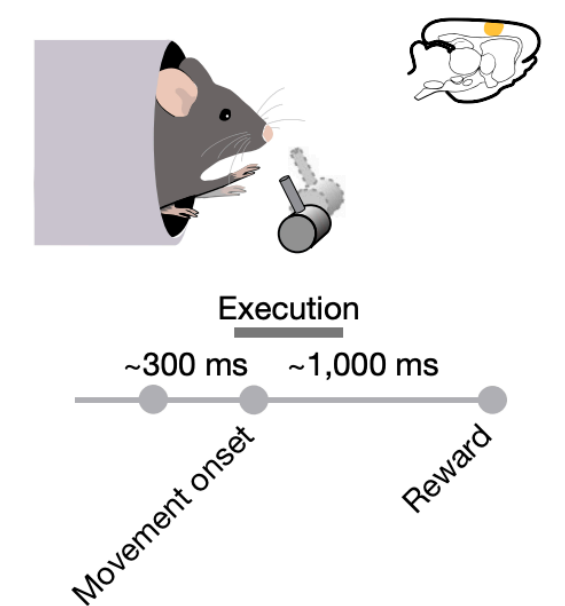
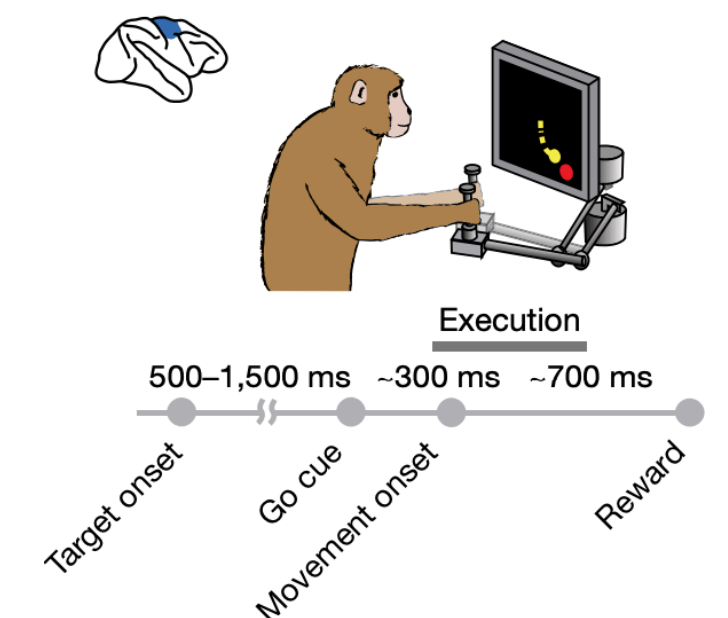
How different covariates impact their inner representations?

Different systems

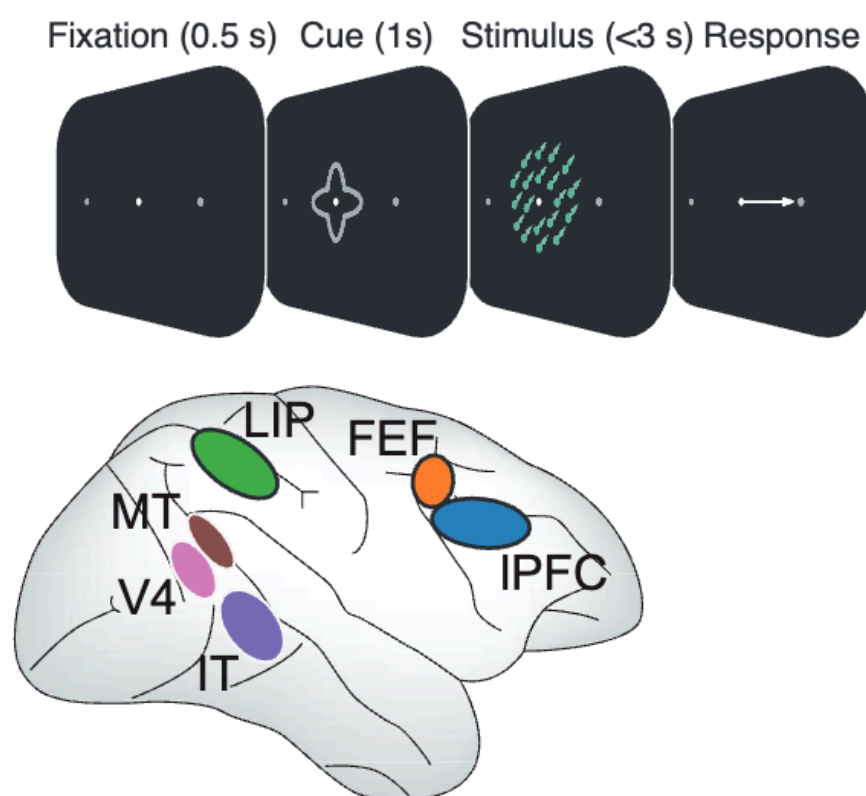
Individuals



Species



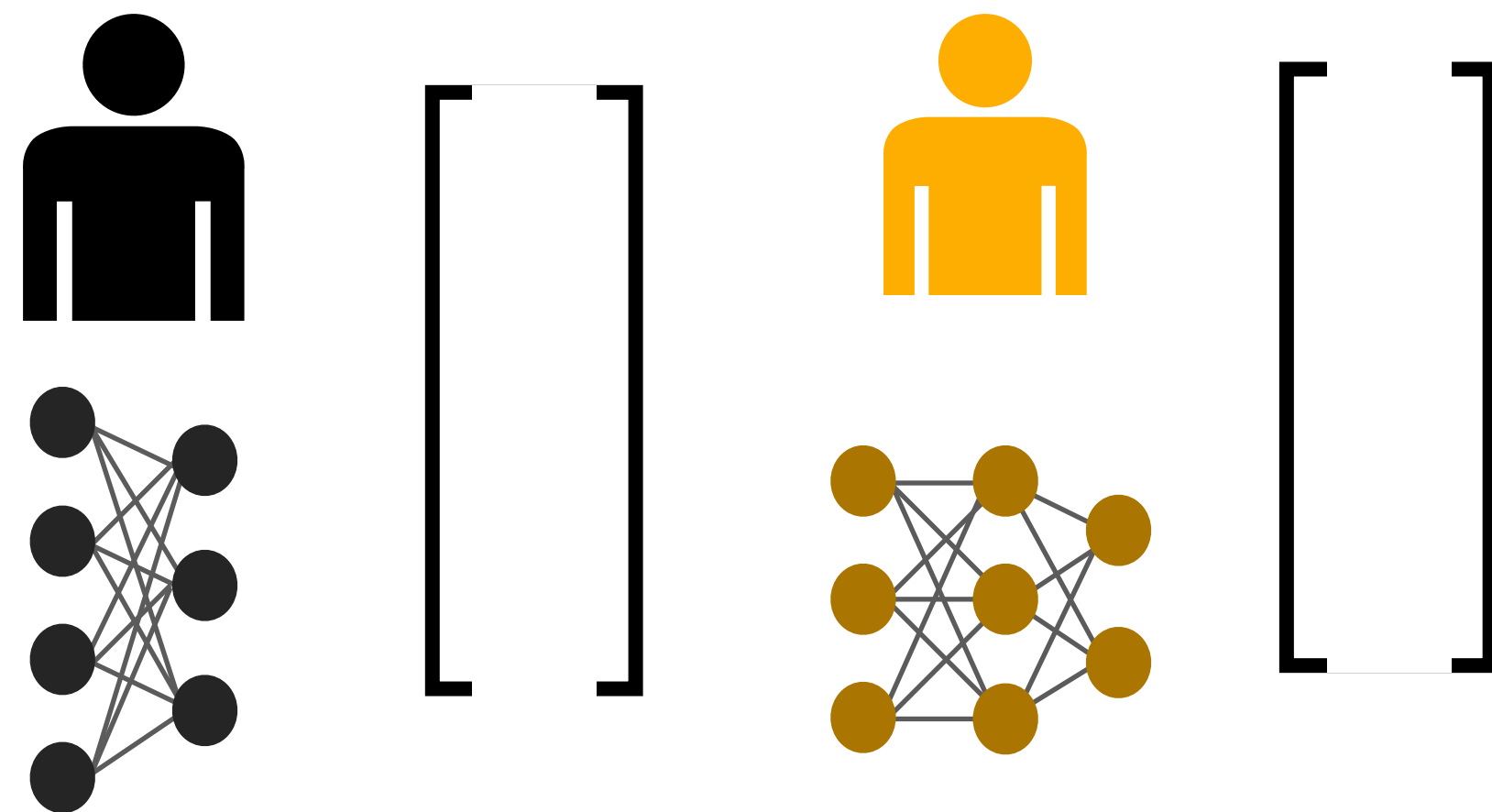
Brain regions



How to **understand** these systems?

For answering this question, you need to

1. Have a proper way to **represent** mathematically these systems

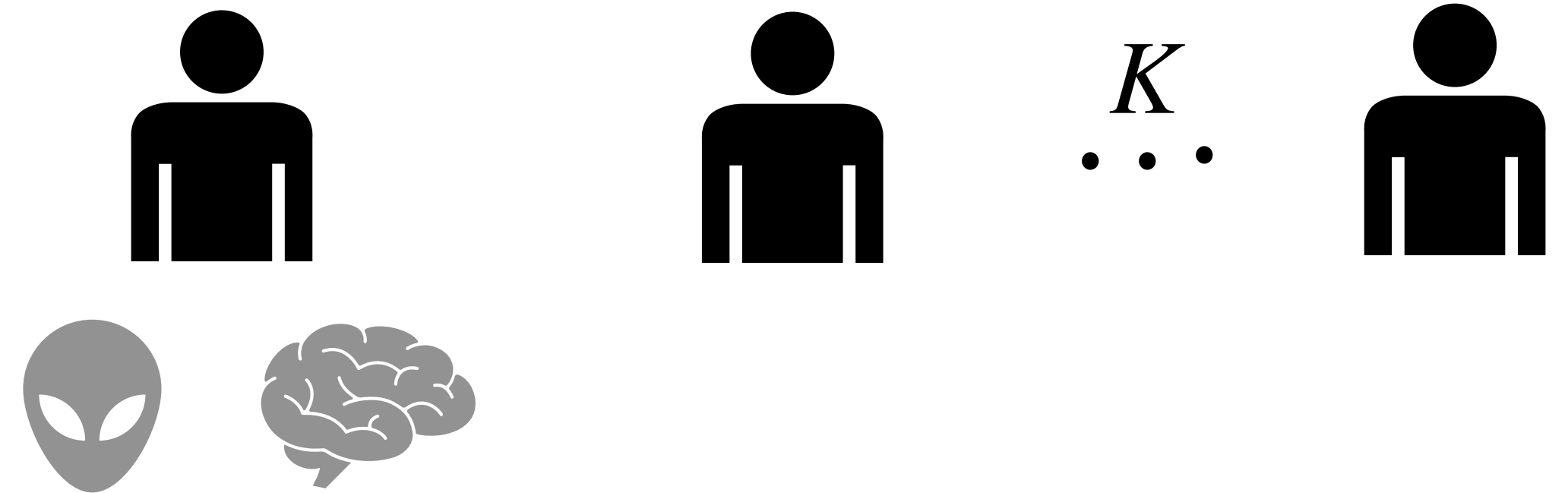


2. Have suited metrics for comparing these objects

$$d\left(\begin{bmatrix} \end{bmatrix}, \begin{bmatrix} \end{bmatrix}\right) = ?$$

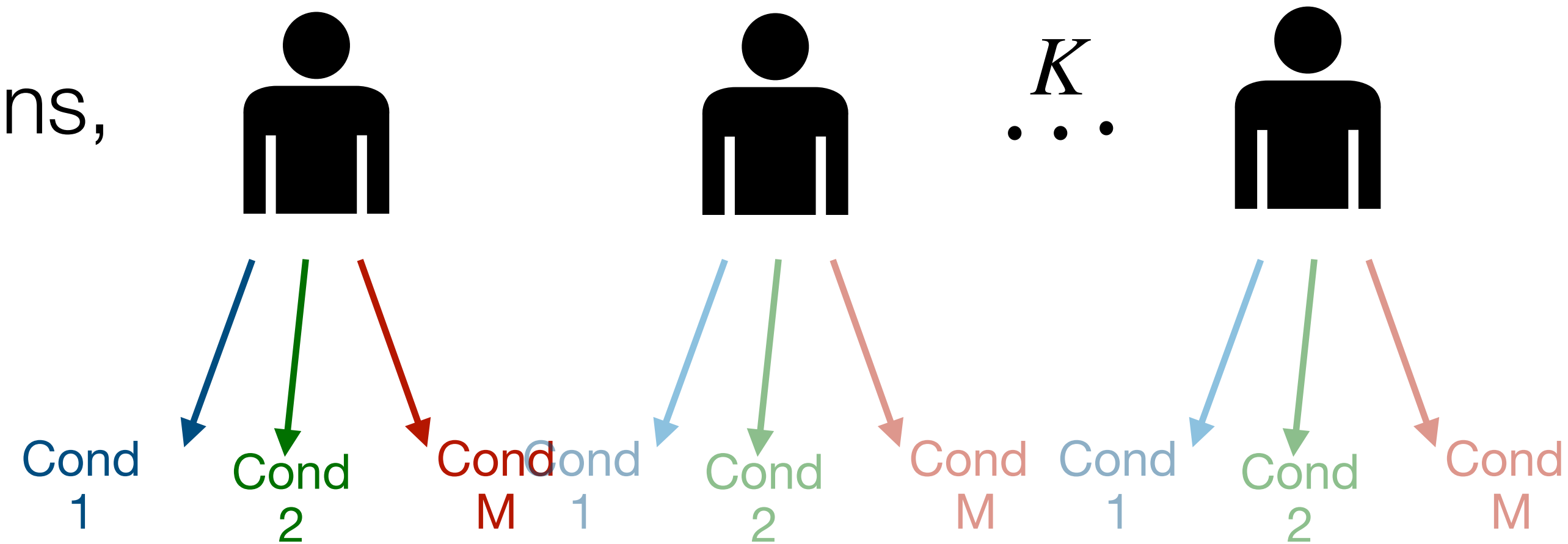
Representing neural systems

- ▶ Consider K systems (animals, sessions, regions..)



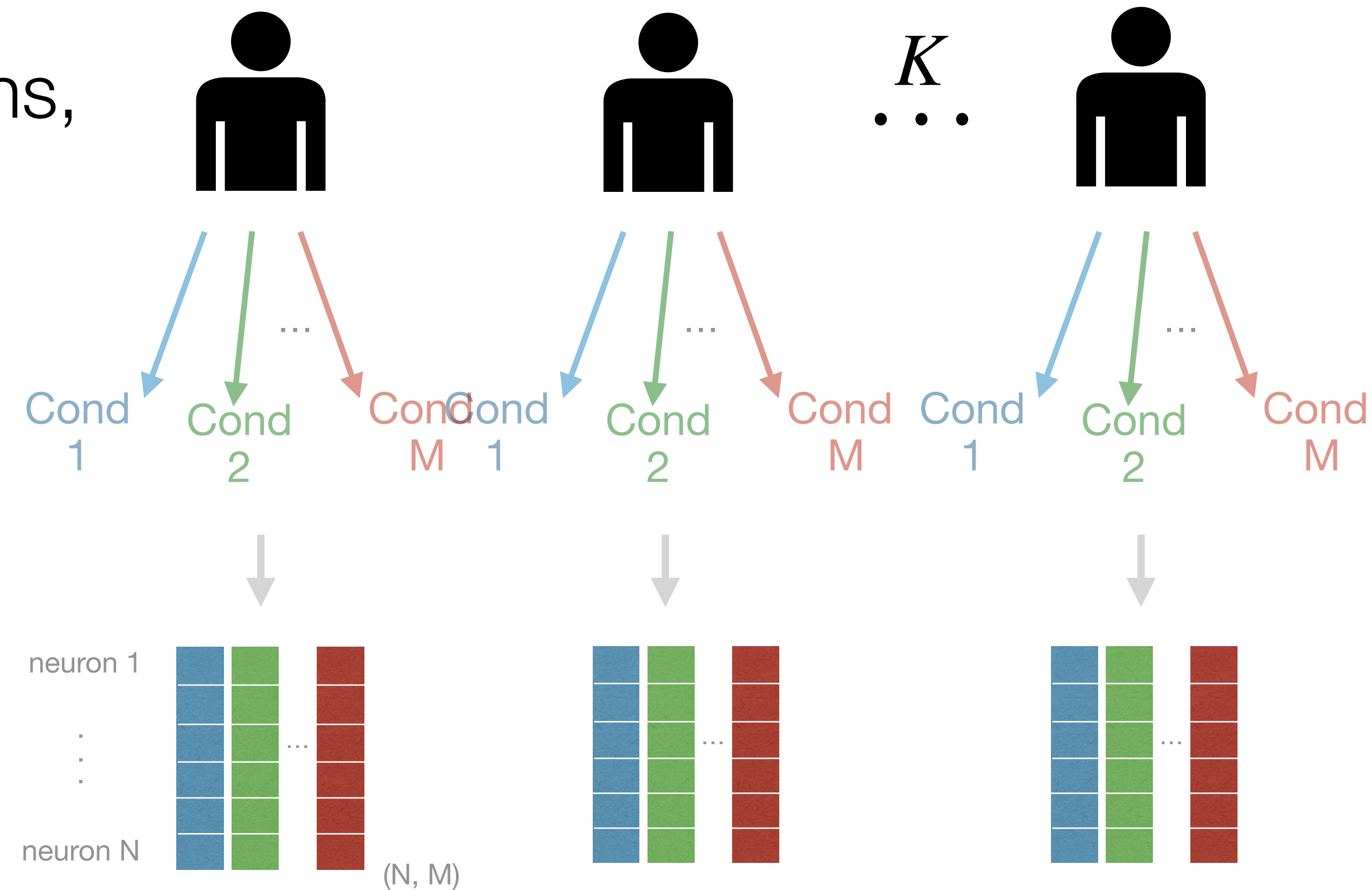
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- ▶ Each subjected to M different conditions



Representing neural systems

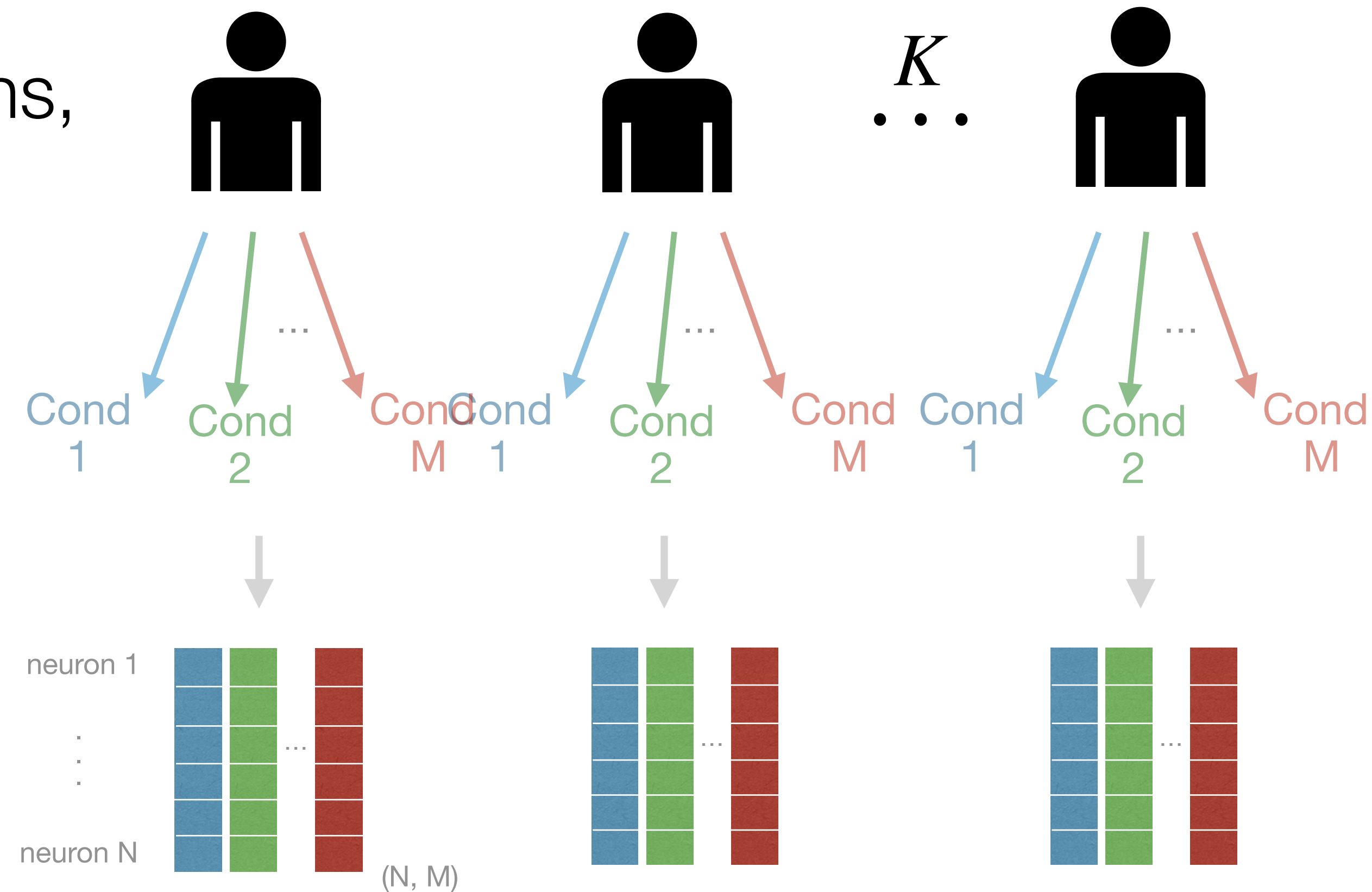
- ▶ Consider K systems (animals, sessions, regions..)
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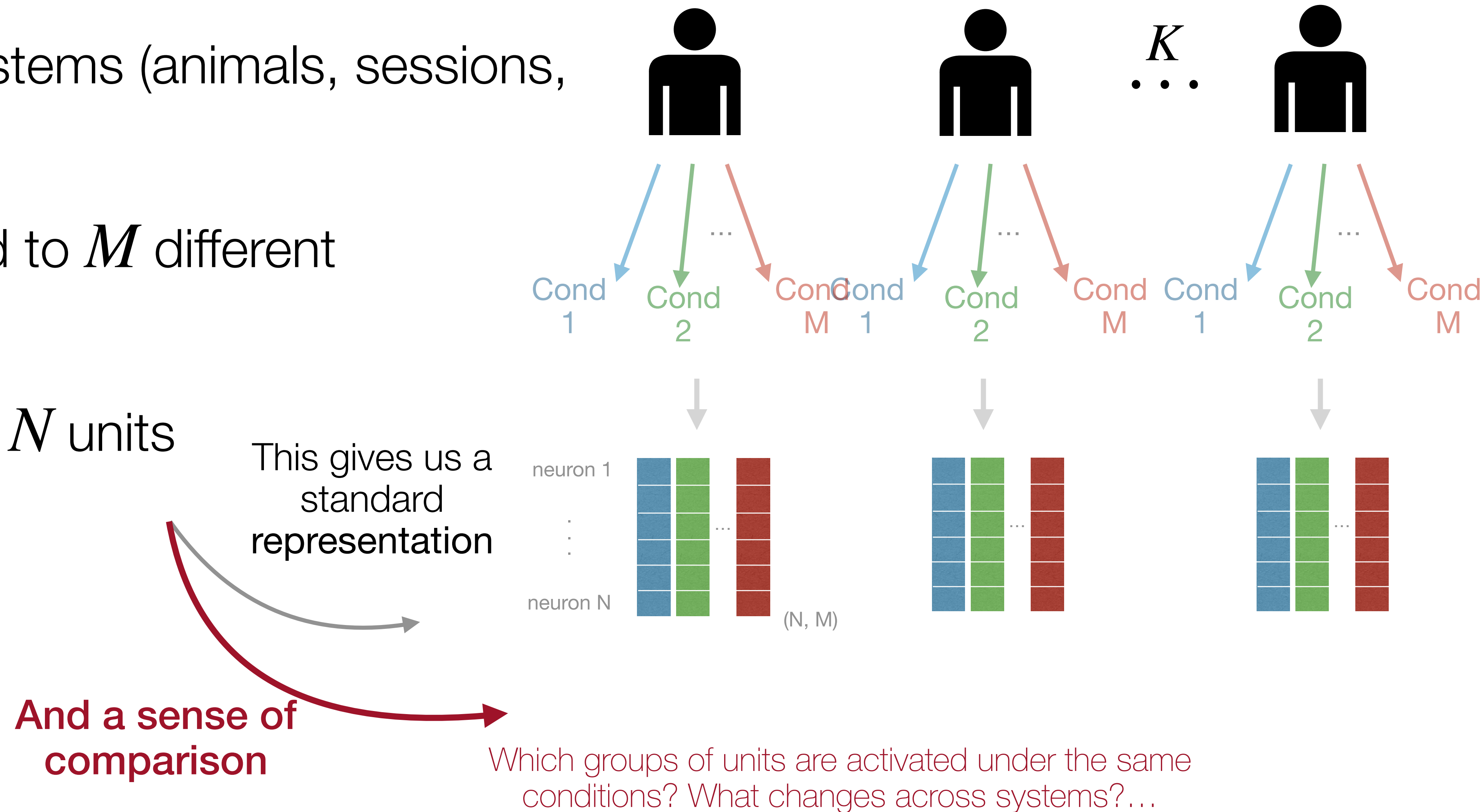
This gives us a
standard
representation



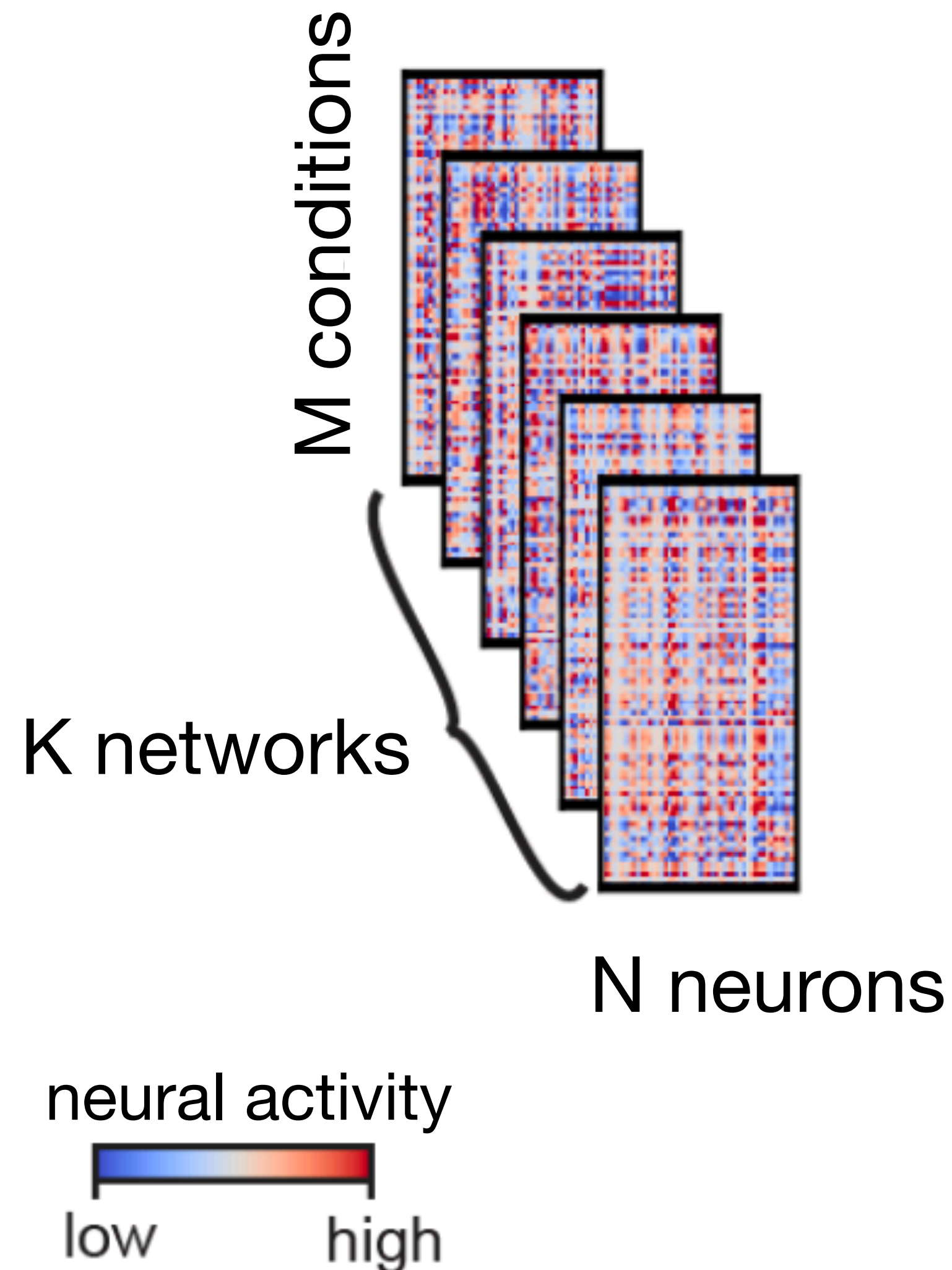
Activation of different
features (units)
under multiple stimulus

Representing neural systems

- ▶ Consider K systems (animals, sessions, regions..)
- ▶ Each subjected to M different conditions
- ▶ Recorded over N units



Distance metrics **enable comparison**



- ▶ Visualize all K networks in a low-D space
- ▶ Find clusters of similar networks
- ▶ Predict network behavior from nearest neighbors
- ▶ Compute “average” geometry across K networks
- ▶ Compute variance around this average geometry

→ *To what extent do the systems “agree”?*

Selecting a distance measure

A **metric** is a function $d(\mathbf{X}, \mathbf{Y})$ which satisfies:

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1. $d(\mathbf{X}, \mathbf{Y}) = 0 \iff \mathbf{X} \sim \mathbf{Y}$ *(Equivalence Condition)*

2. $d(\mathbf{X}, \mathbf{Y}) = d(\mathbf{Y}, \mathbf{X})$ *(Symmetry)*

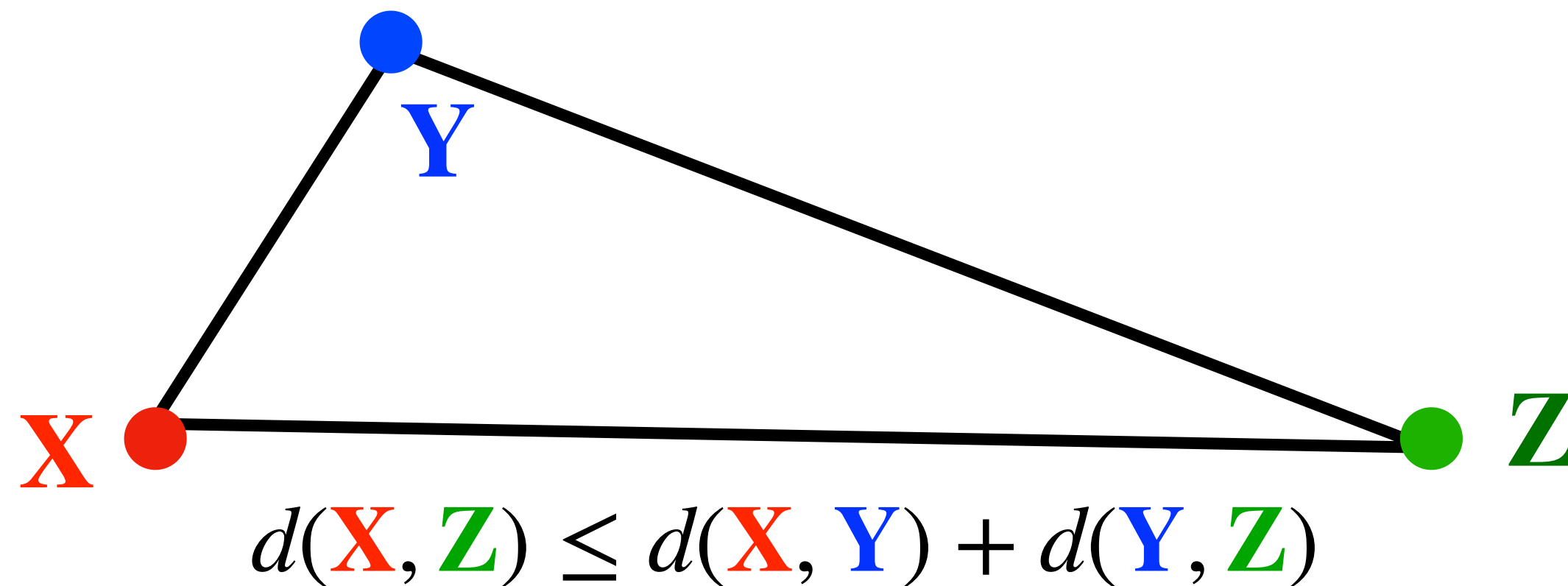
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2. $d(\mathbf{X}, \mathbf{Y}) = d(\mathbf{Y}, \mathbf{X})$ *(Symmetry)*

3. $d(\mathbf{X}, \mathbf{Z}) \leq d(\mathbf{X}, \mathbf{Y}) + d(\mathbf{Y}, \mathbf{Z})$ *(Triangle Inequality)*



Q: Why define a metric space?

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A: Many analyses (e.g. clustering) only make sense in metric spaces.

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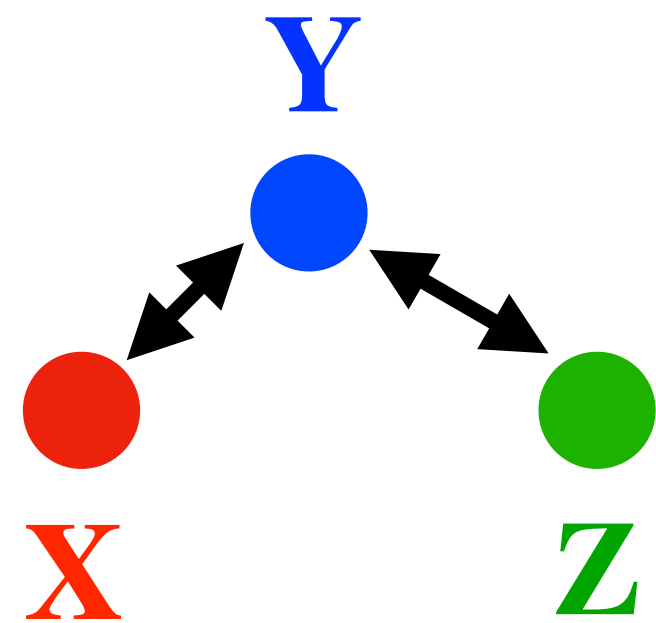
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Suppose the triangle inequality is violated...

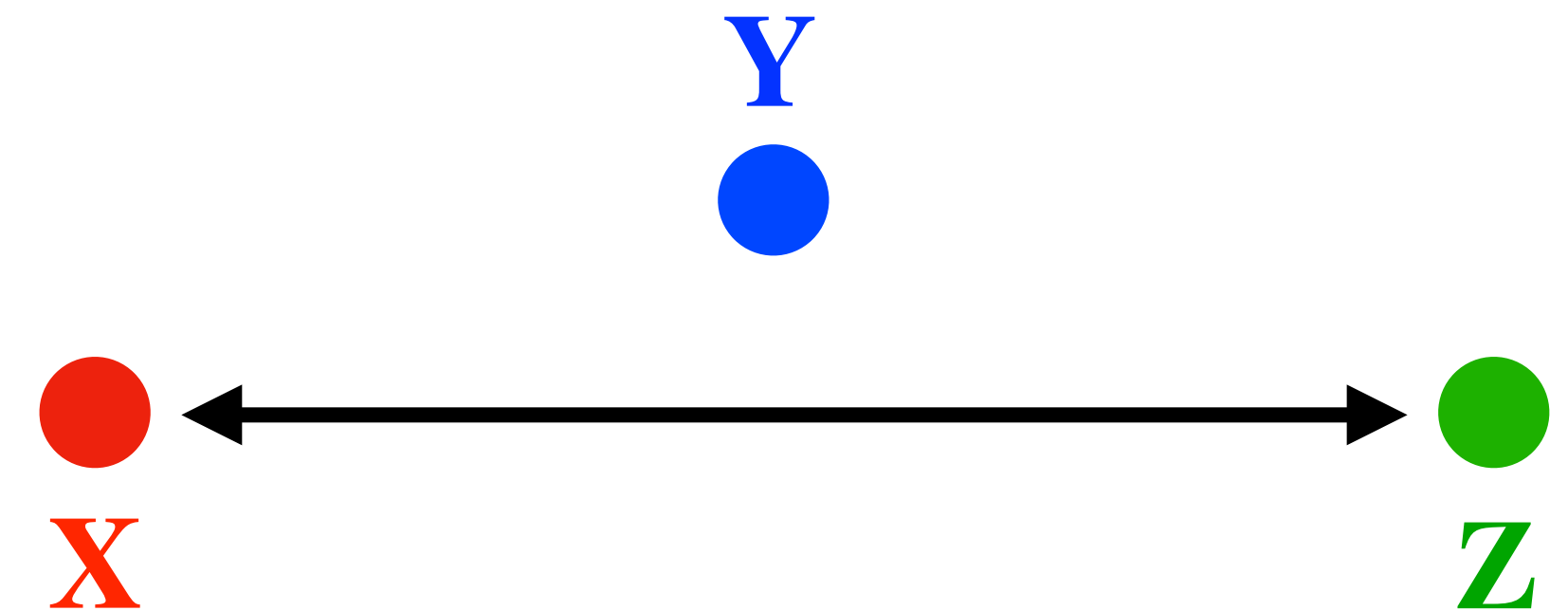
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Suppose the triangle inequality is violated...



but at the same time....



X is close to **Y**
Y is close to **Z**

X and **Z** are really far apart

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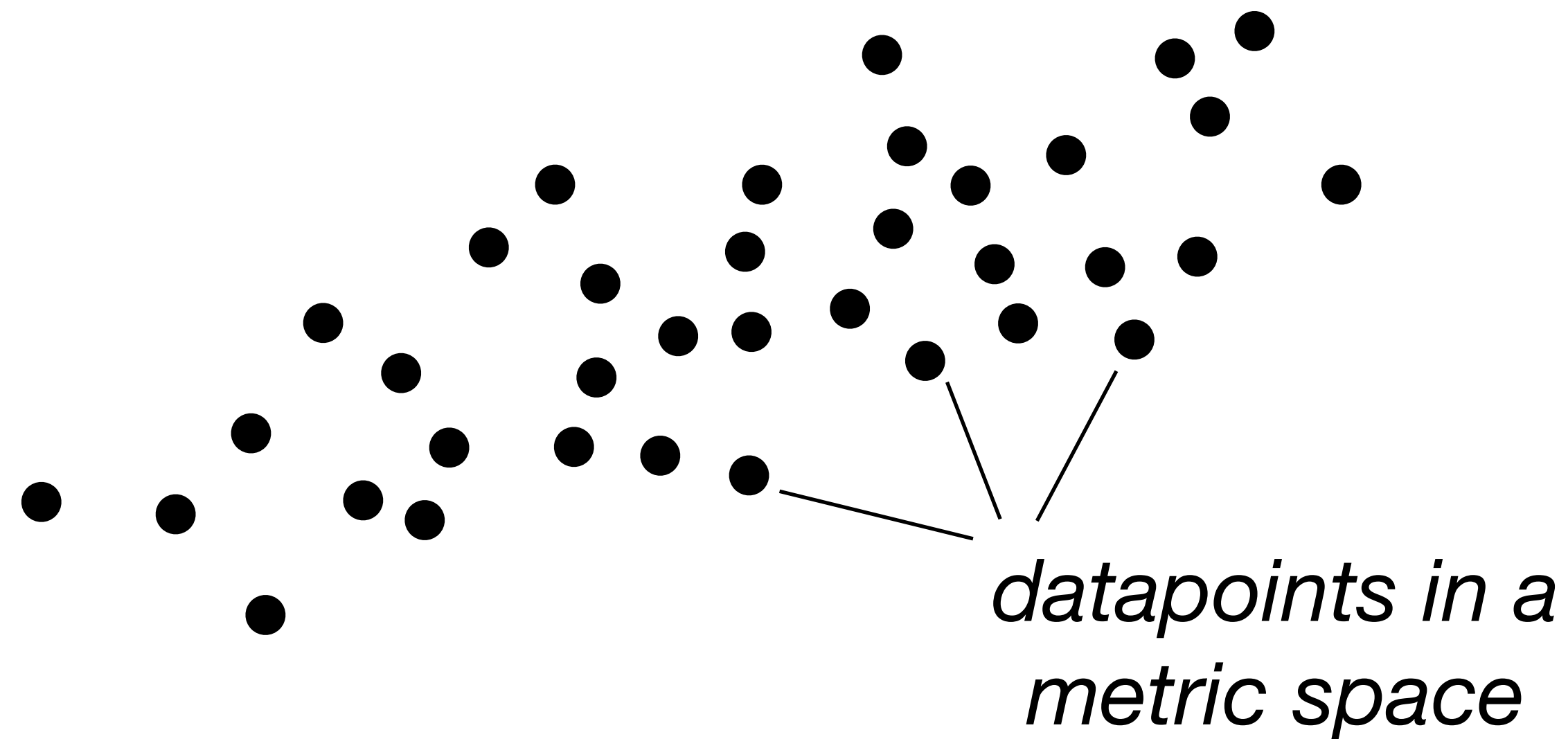
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The triangle inequality makes it easy to define a “neighborhood”
around each datapoint:

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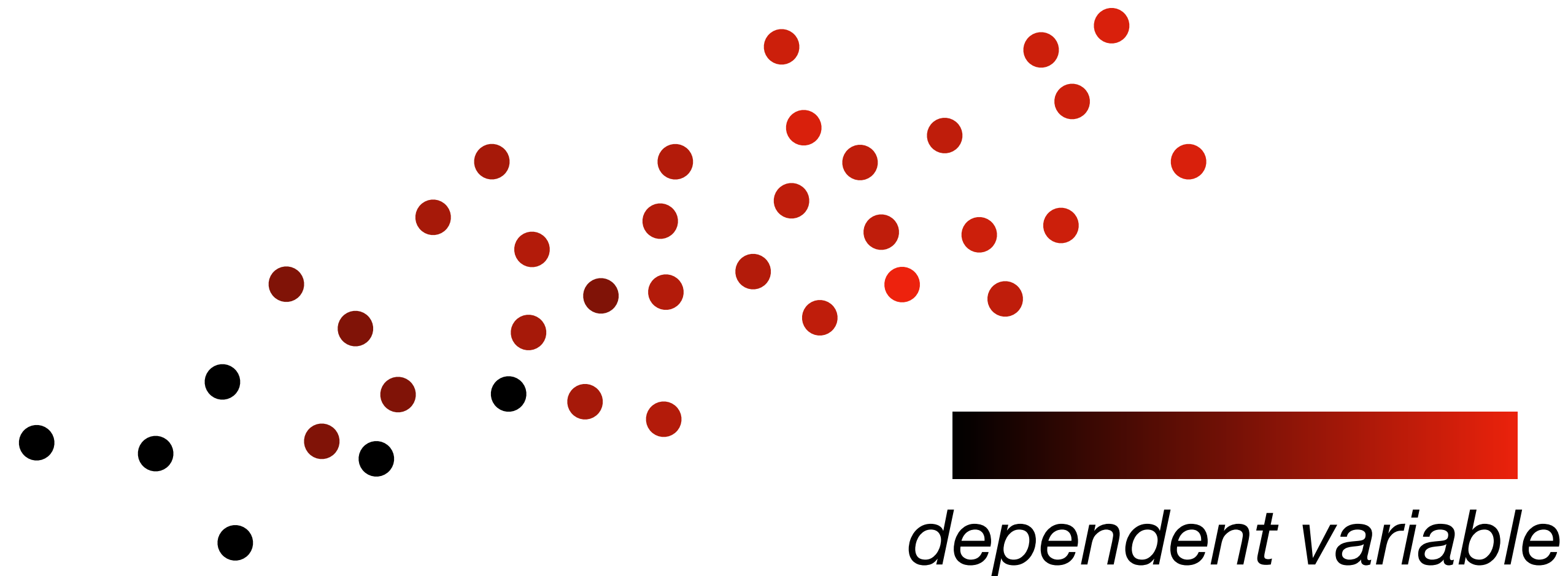
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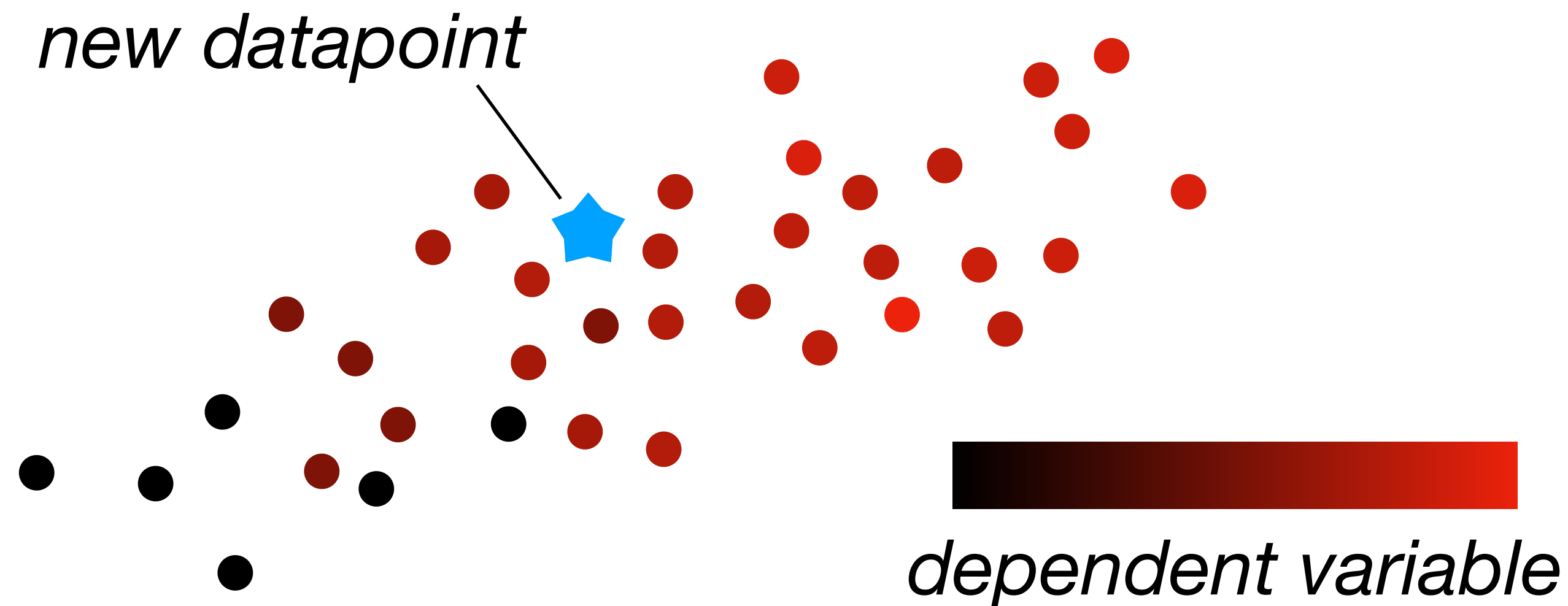
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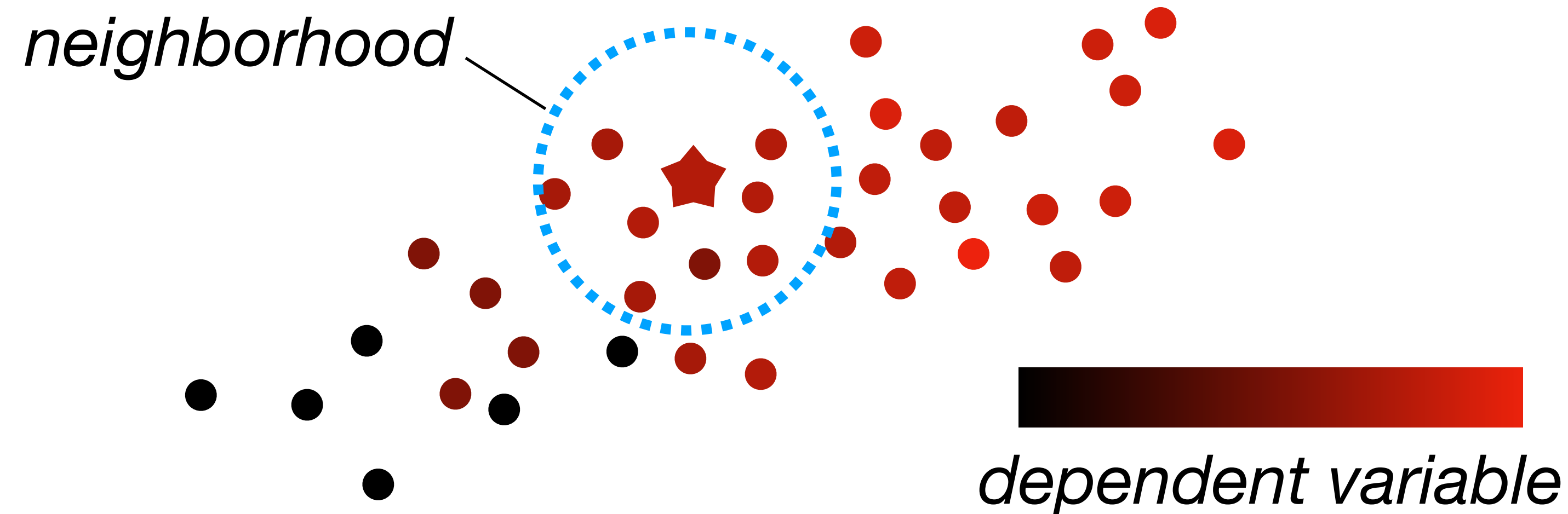
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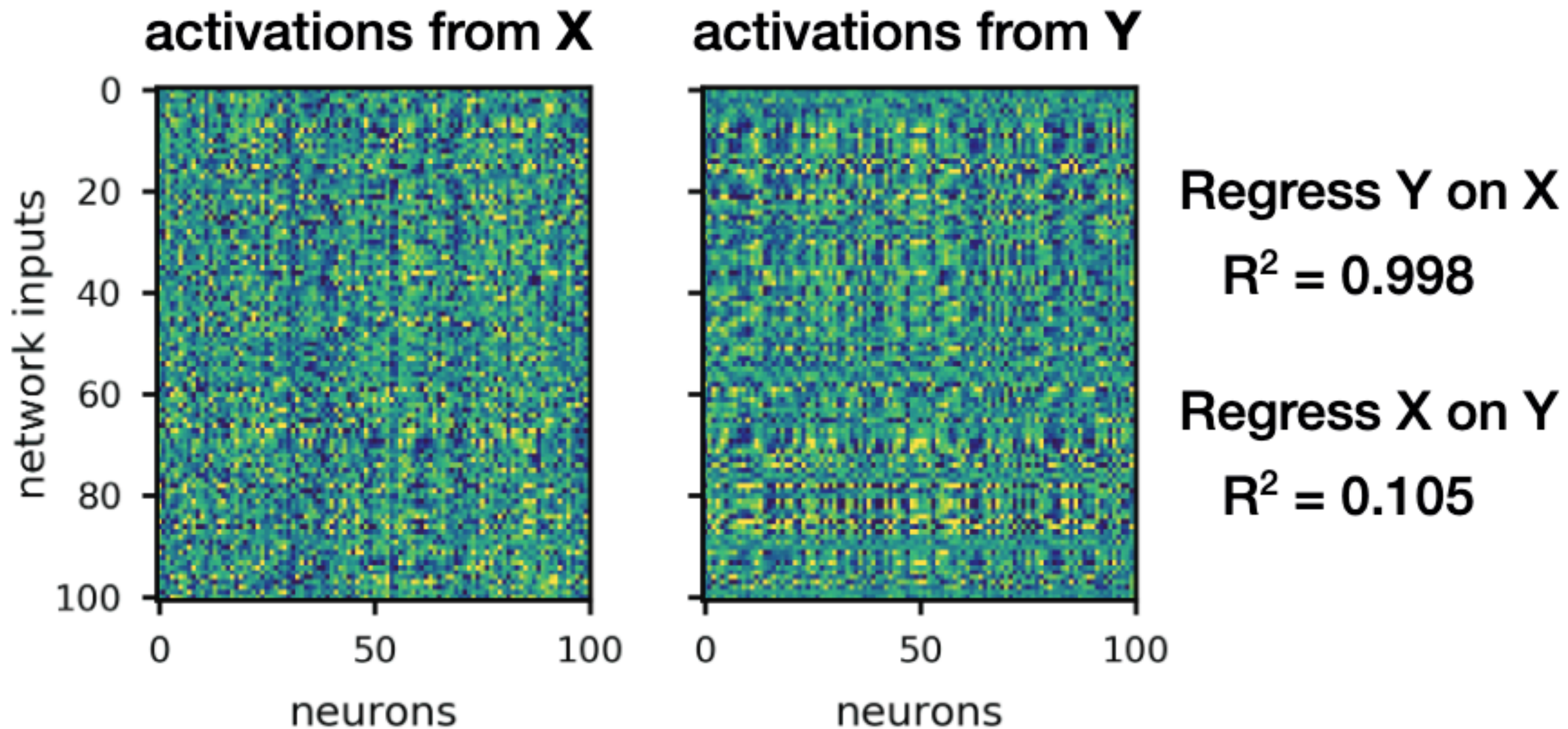
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A: Many don't.

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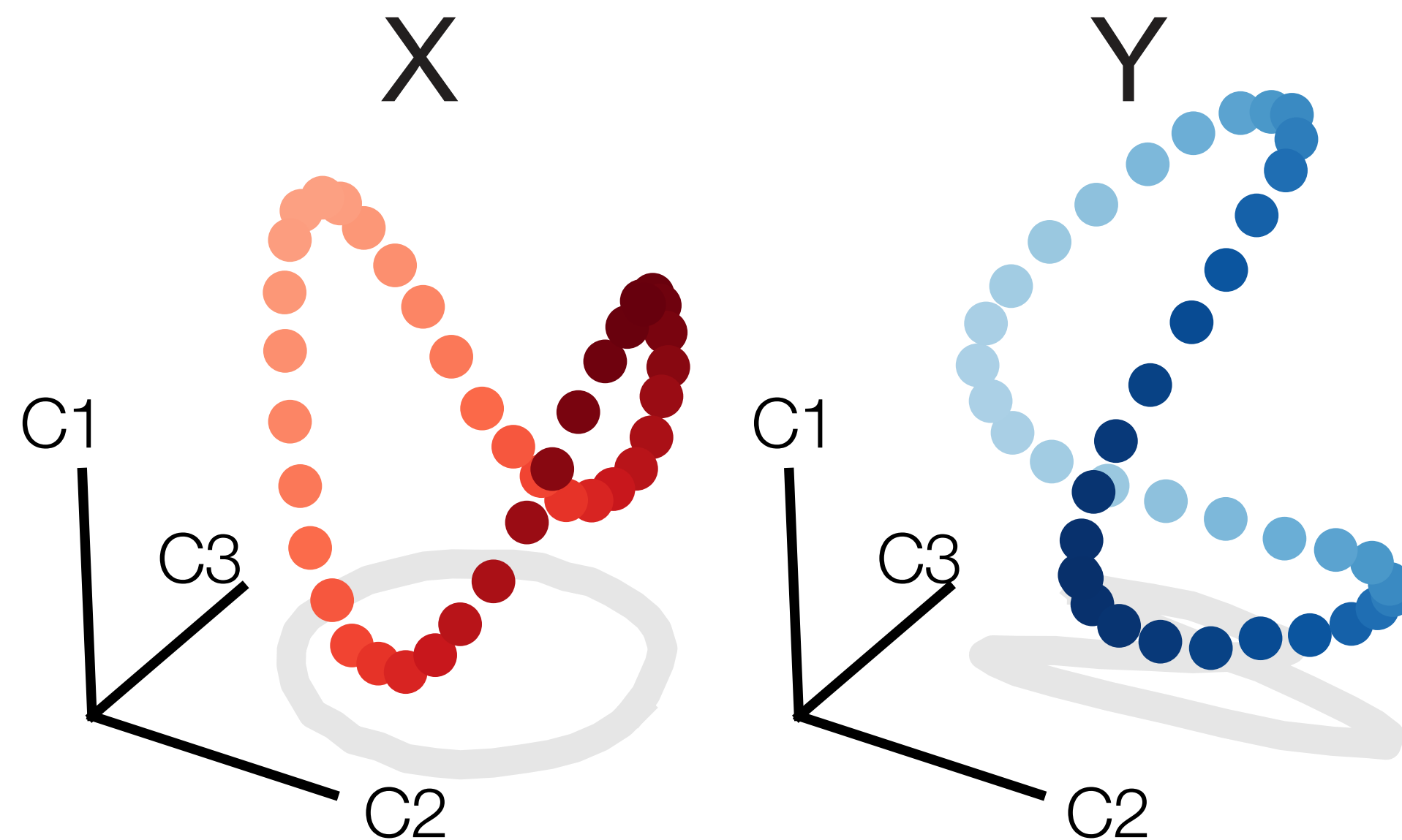
A: Many don't.

Example: Linear Regression / decoding — **Not symmetric!**

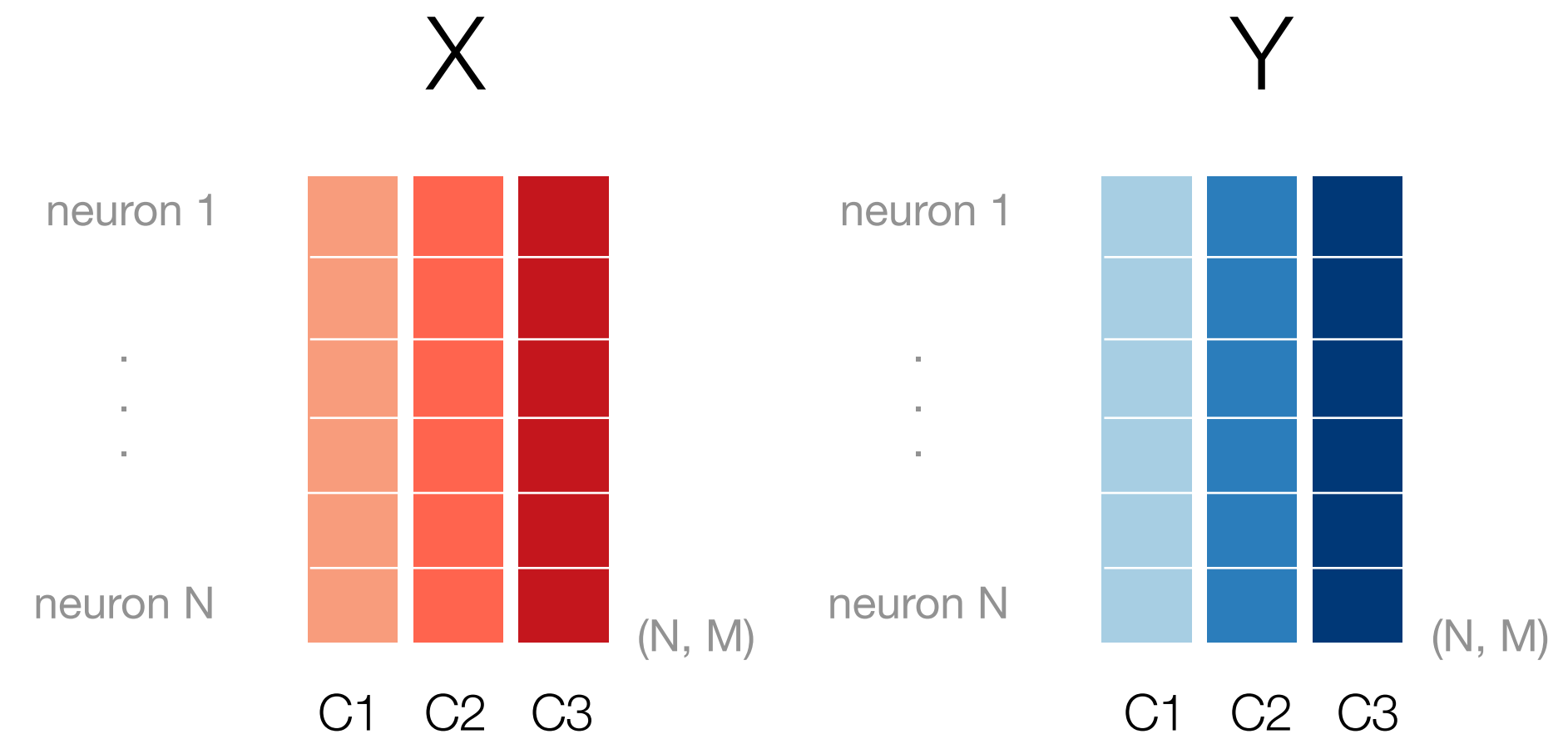


What could be a suited distance?

Let's take 2 sets of points, each defining a shape in a space

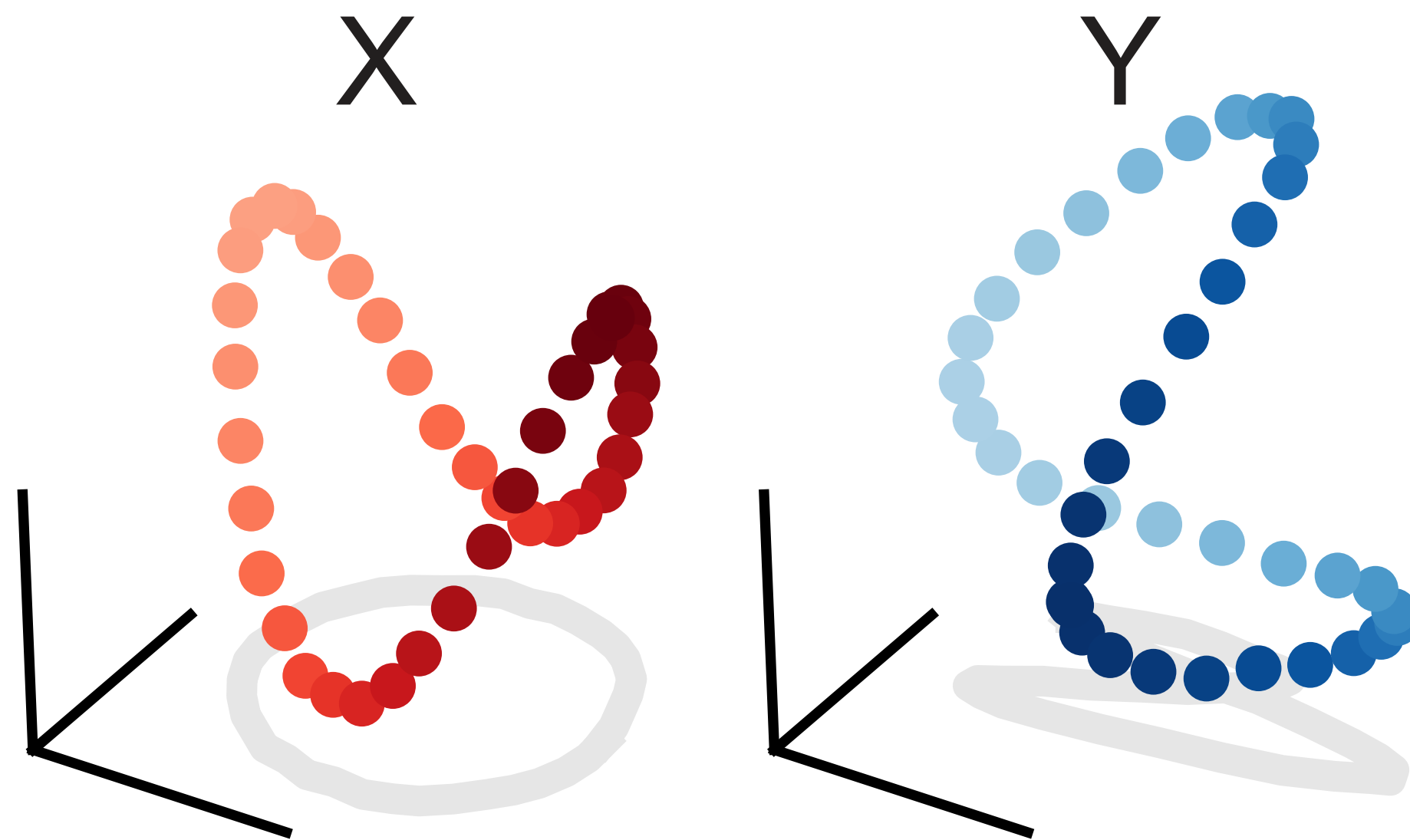


Thinking of the (neurons, conditions) matrices, each dot could be a neuron and each dimension a condition

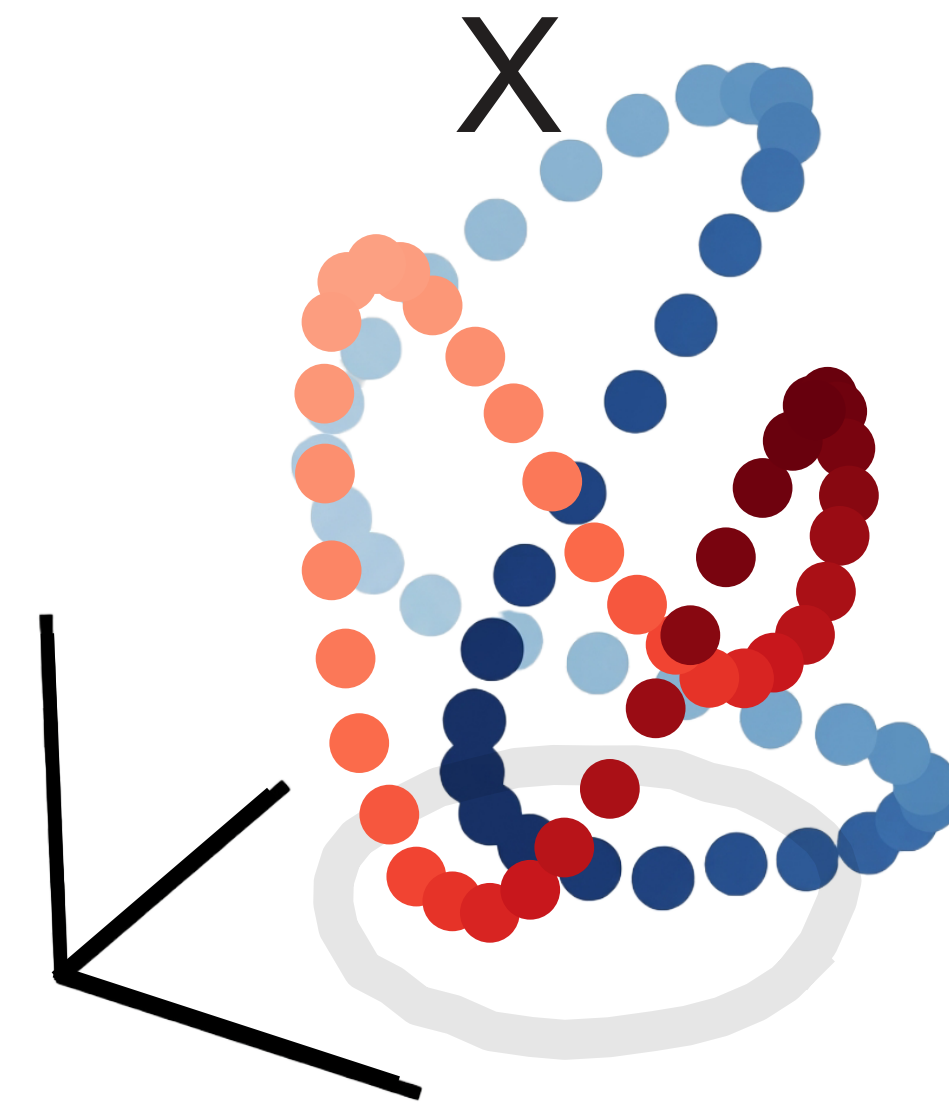


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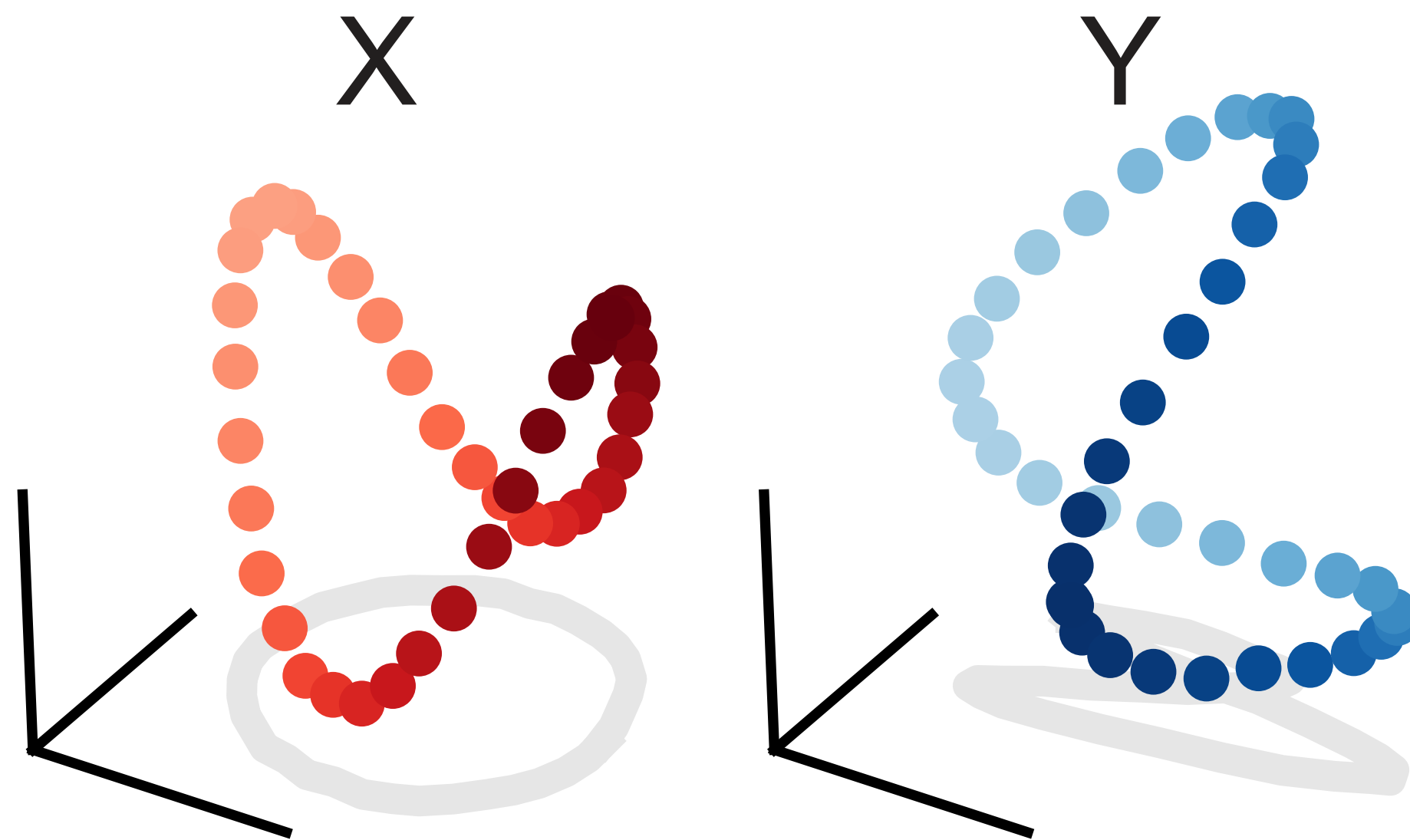


And let them be centred

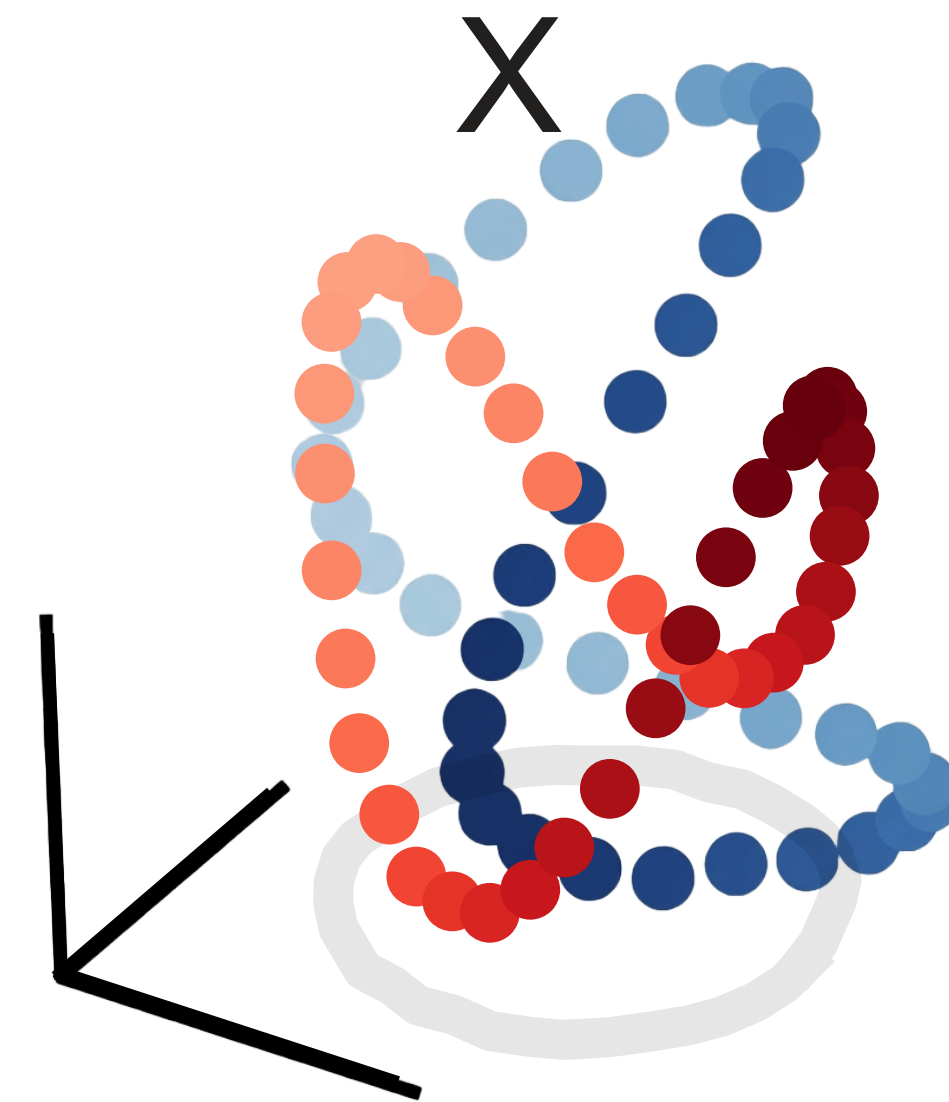


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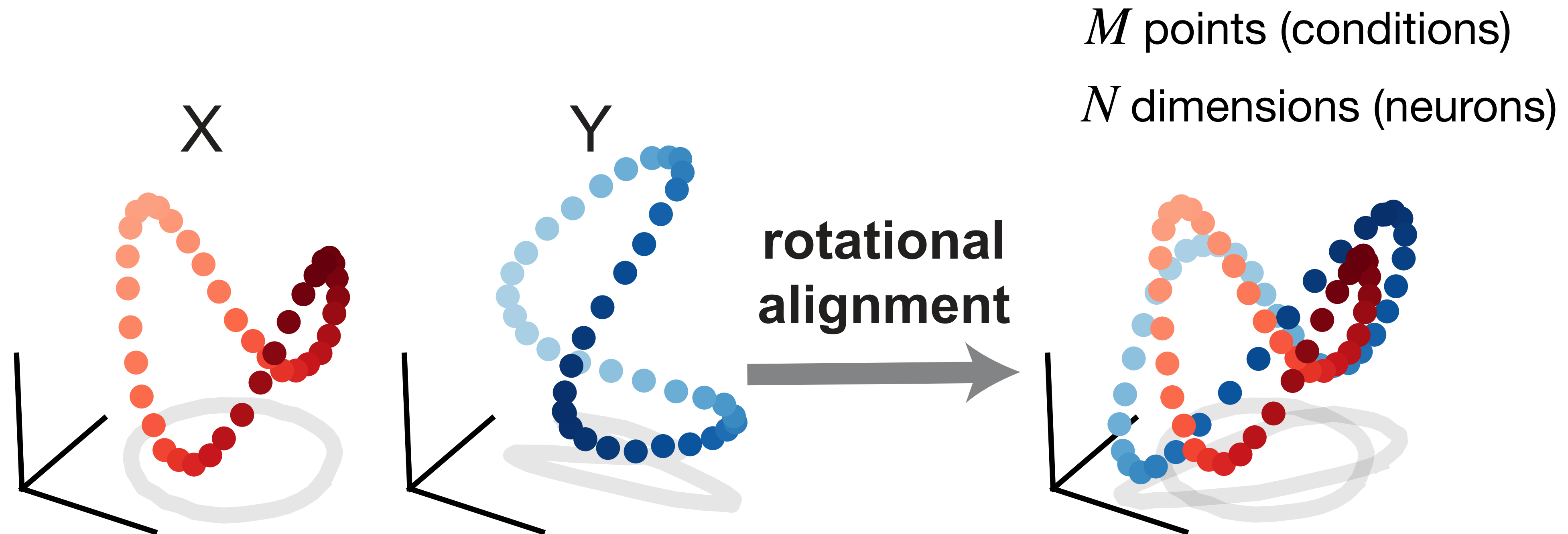
And let them be centred



You need a way to define some distance between
these 2 set of points/shapes
(Not each point individually!)

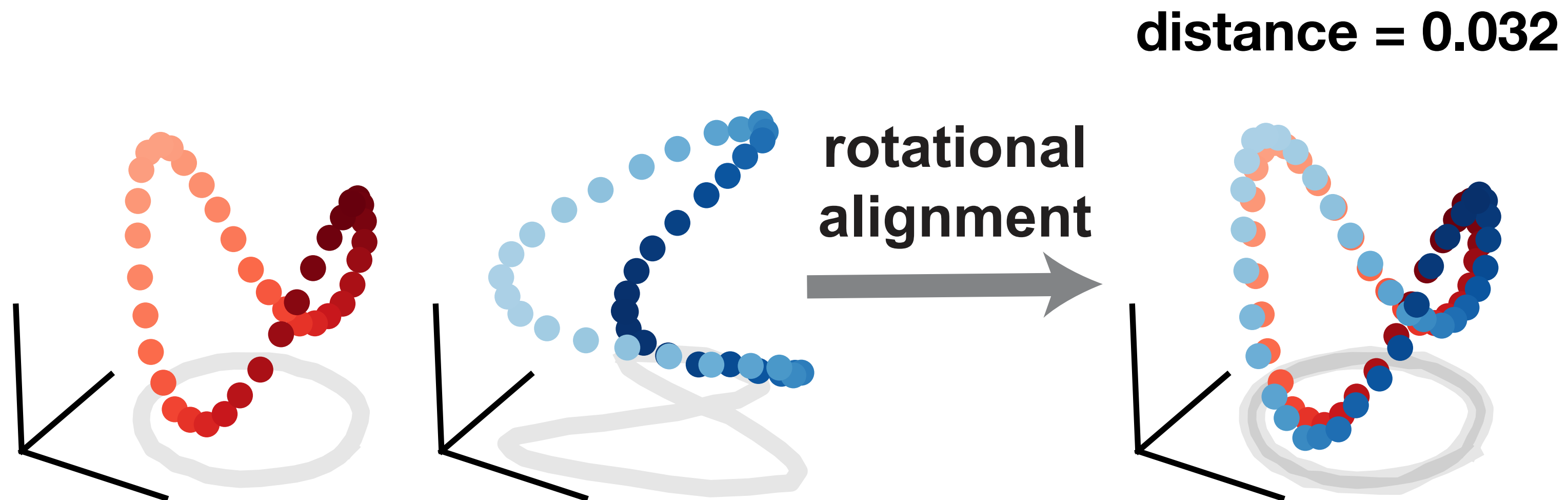
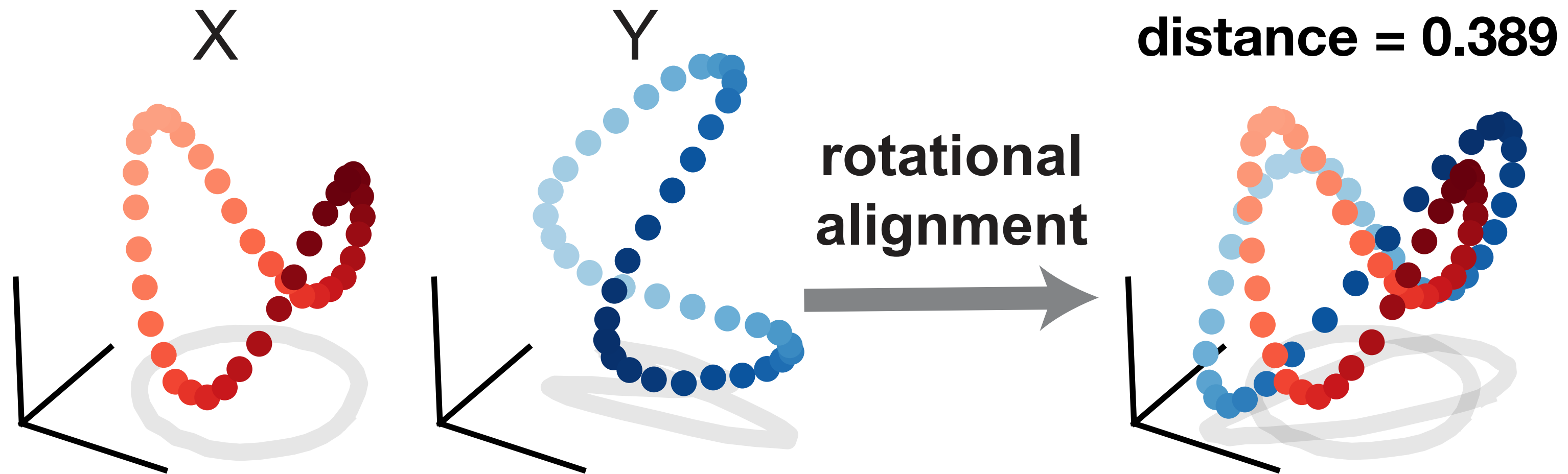
From Procrustes Shape analysis

Distance between 2 shapes is the difference of the coordinates after they are optimally superposed



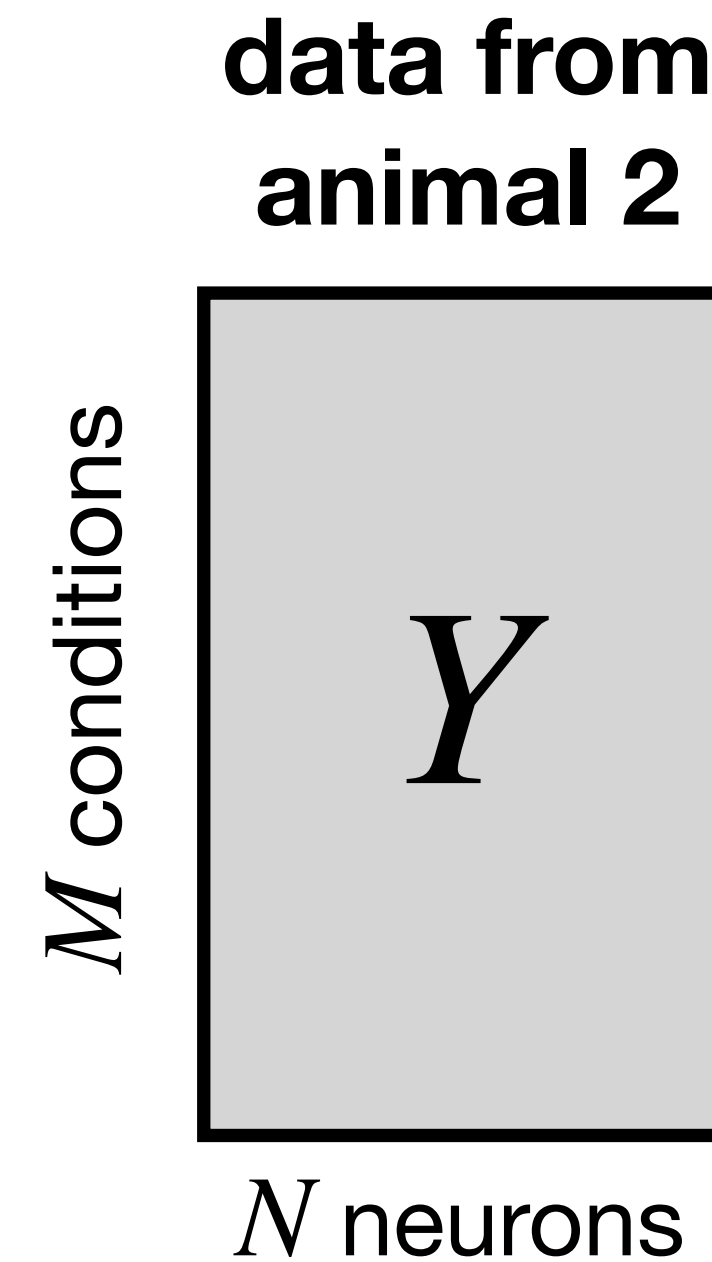
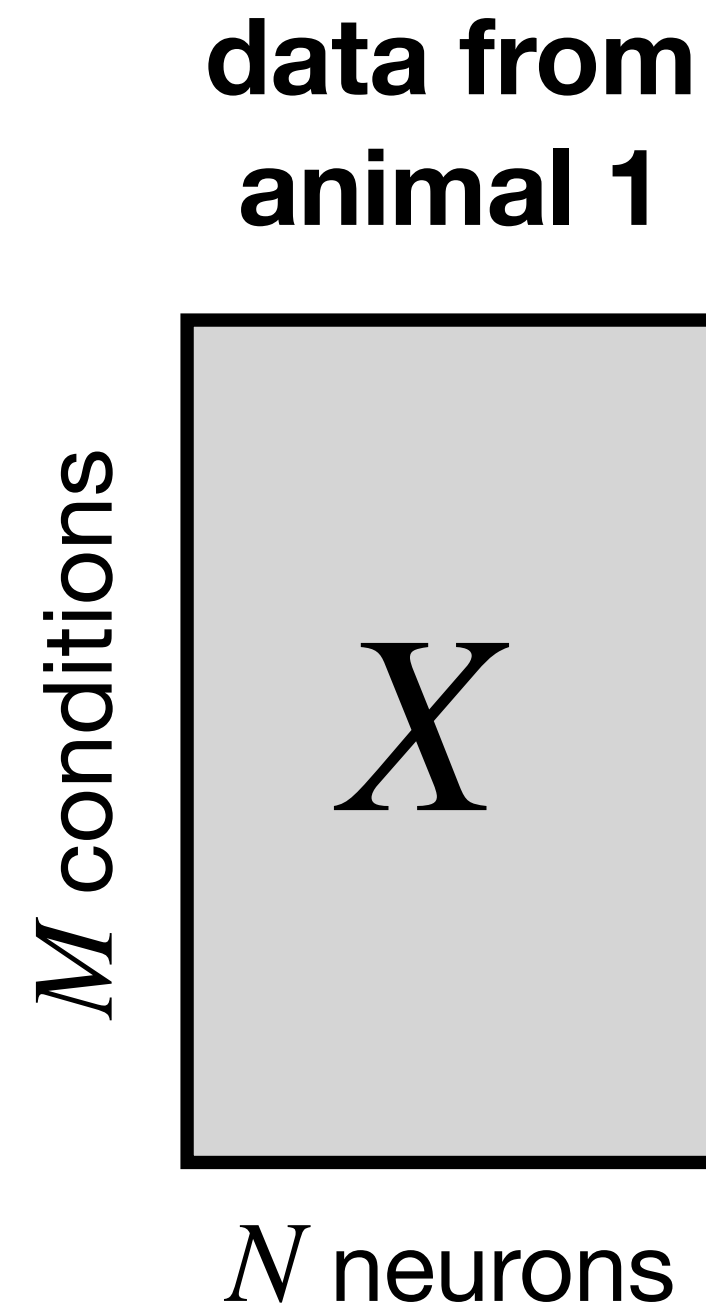
Distance: the magnitude of the residual $X - Y'$ matrix

Procrustes Shape Distance



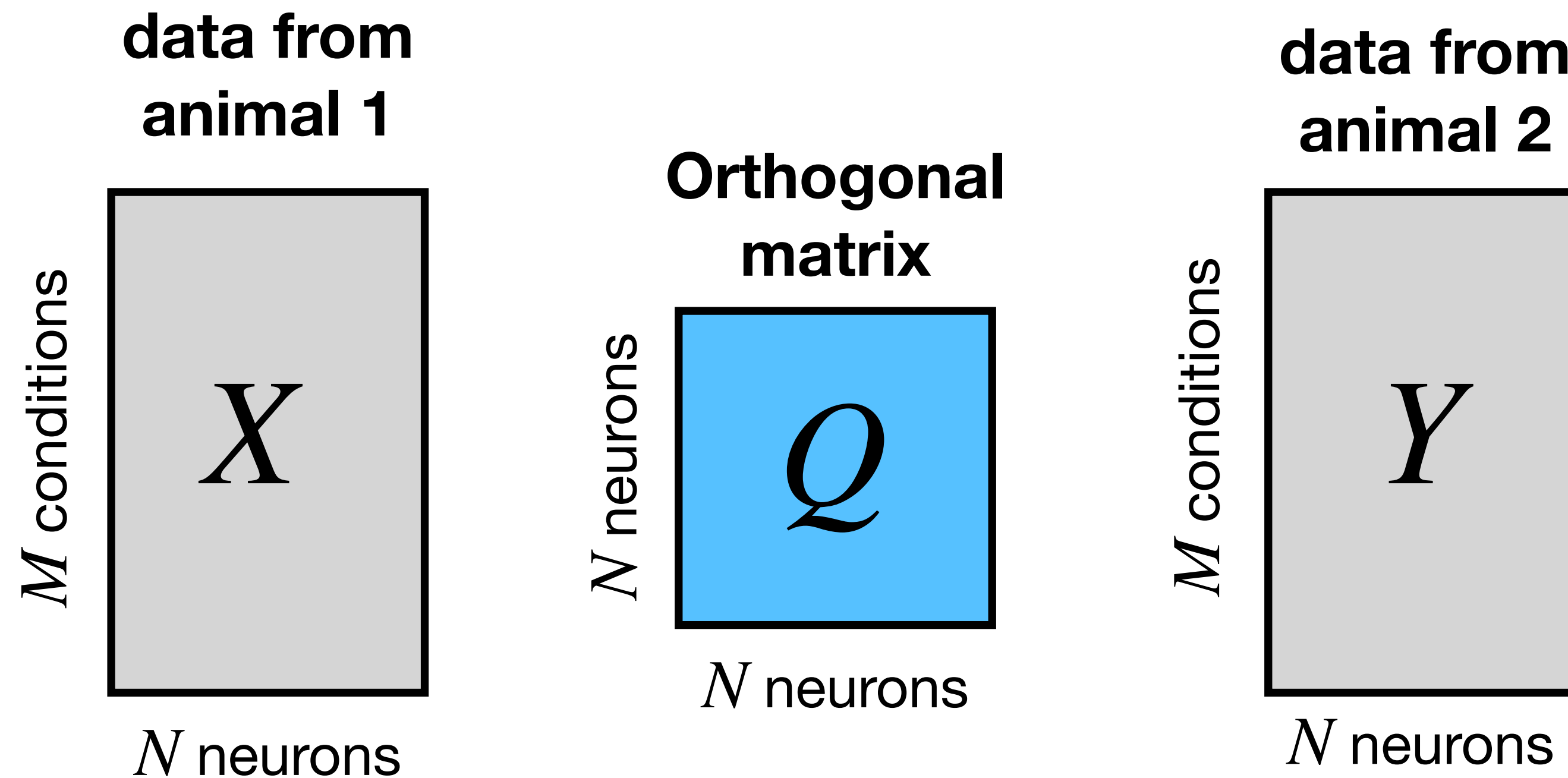
Procrustes Shape Distance

A (constrained) linear prediction score that is a proper metric!



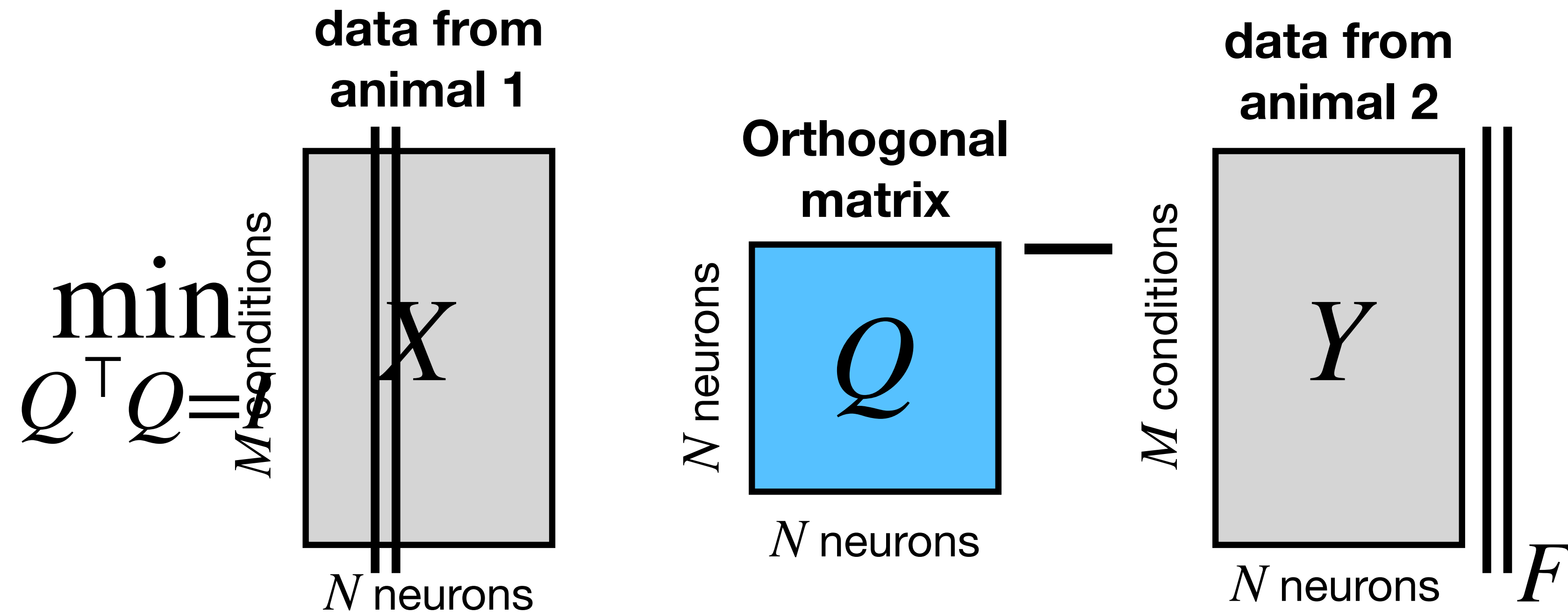
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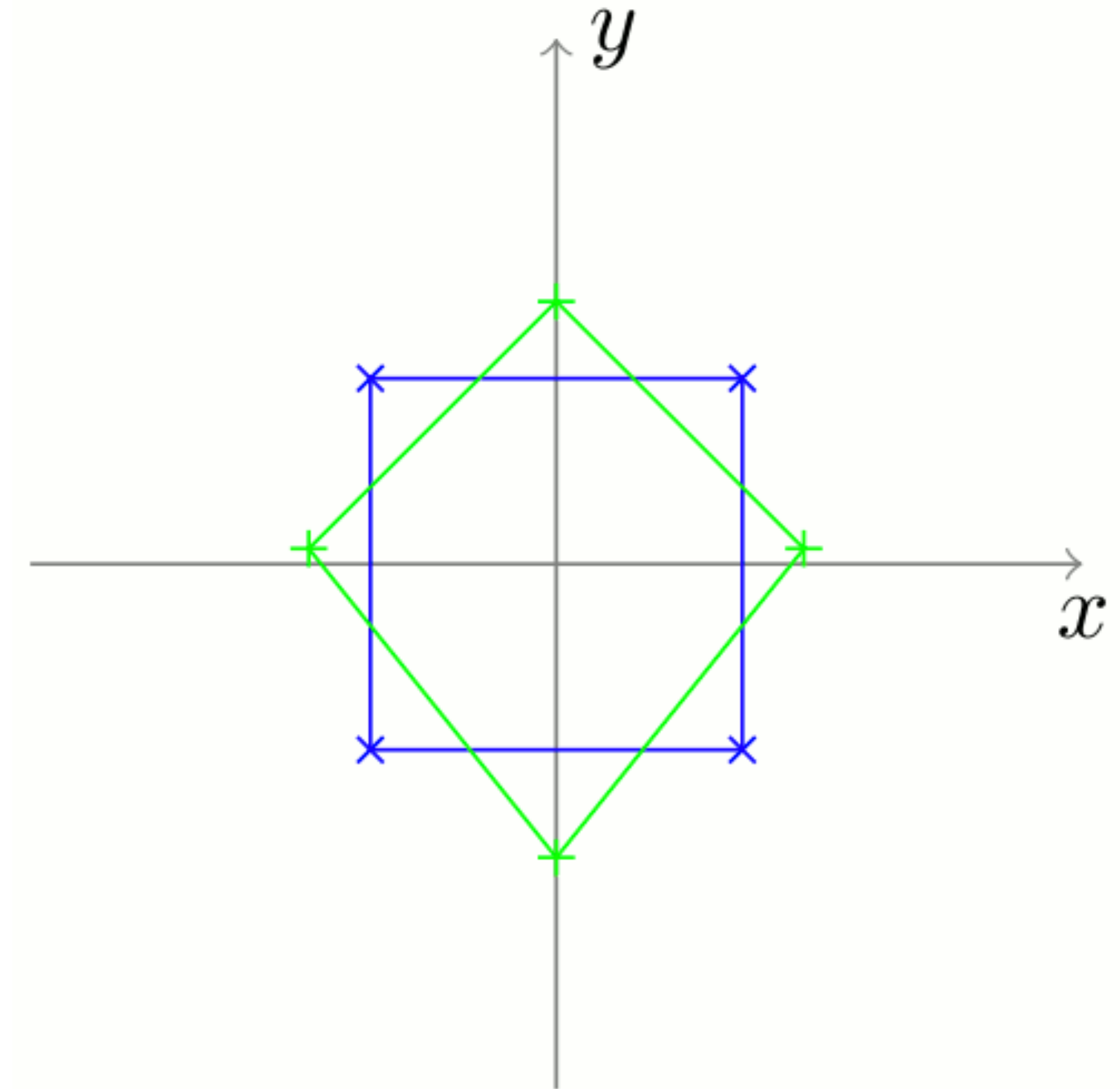
So, to find a distance between your shapes
/ representation matrices, you can

(1) Solve for $\mathbf{Q}$ and find optimal rotation

$$\mathbf{Q}^* = \arg \min_{\mathbf{Q}\mathbf{Q}^T = \mathbf{I}} \|\mathbf{Q}\mathbf{X}_i - \mathbf{X}_j\|_F$$

(2) Substitute in the equation and get the distance

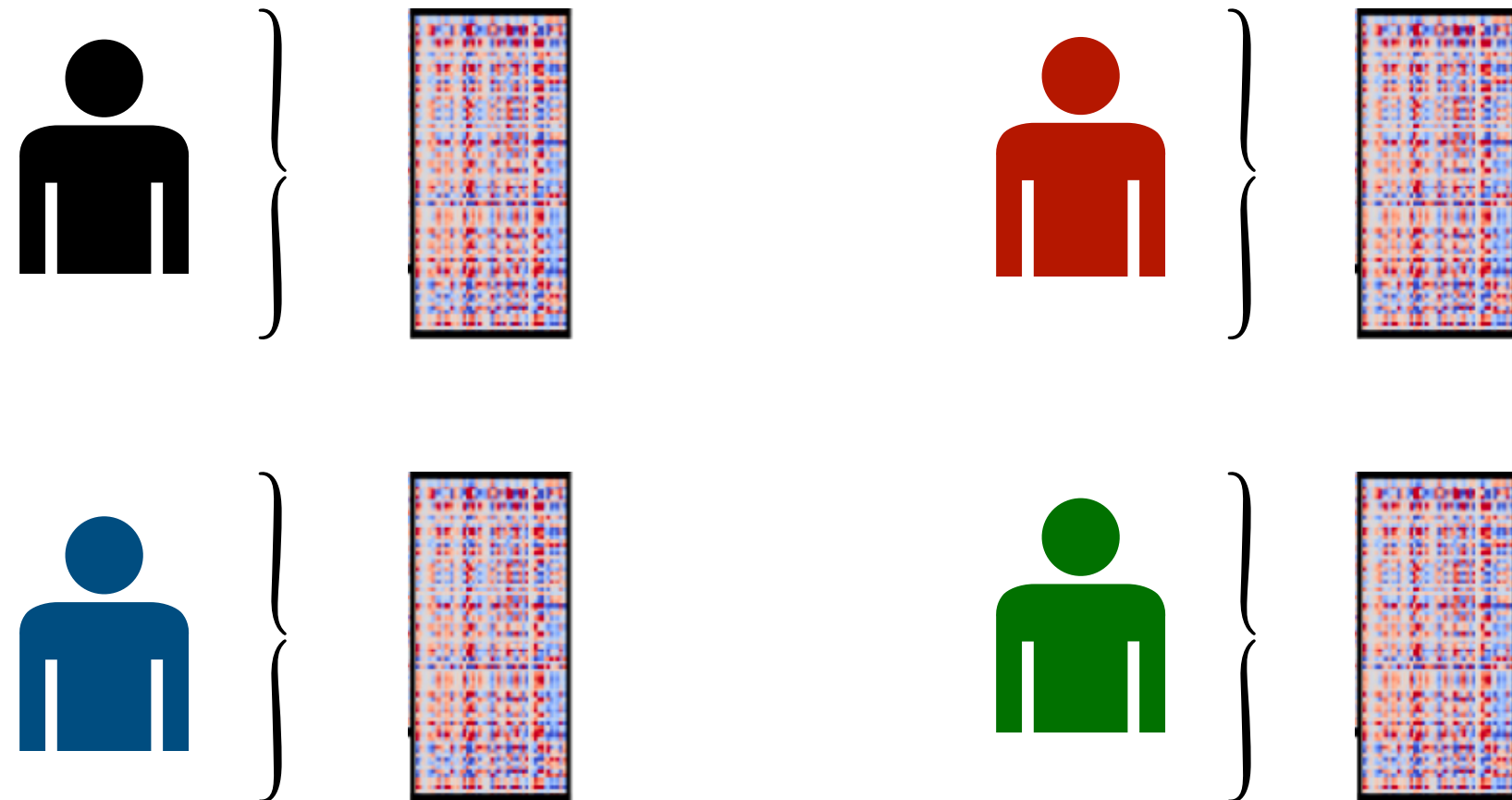
$$d_{\text{Proc}}(\mathbf{X}_i, \mathbf{X}_j) = \|\mathbf{Q}^*\mathbf{X}_i - \mathbf{X}_j\|_F$$



Going back to our initial question:

How to understand complex systems?

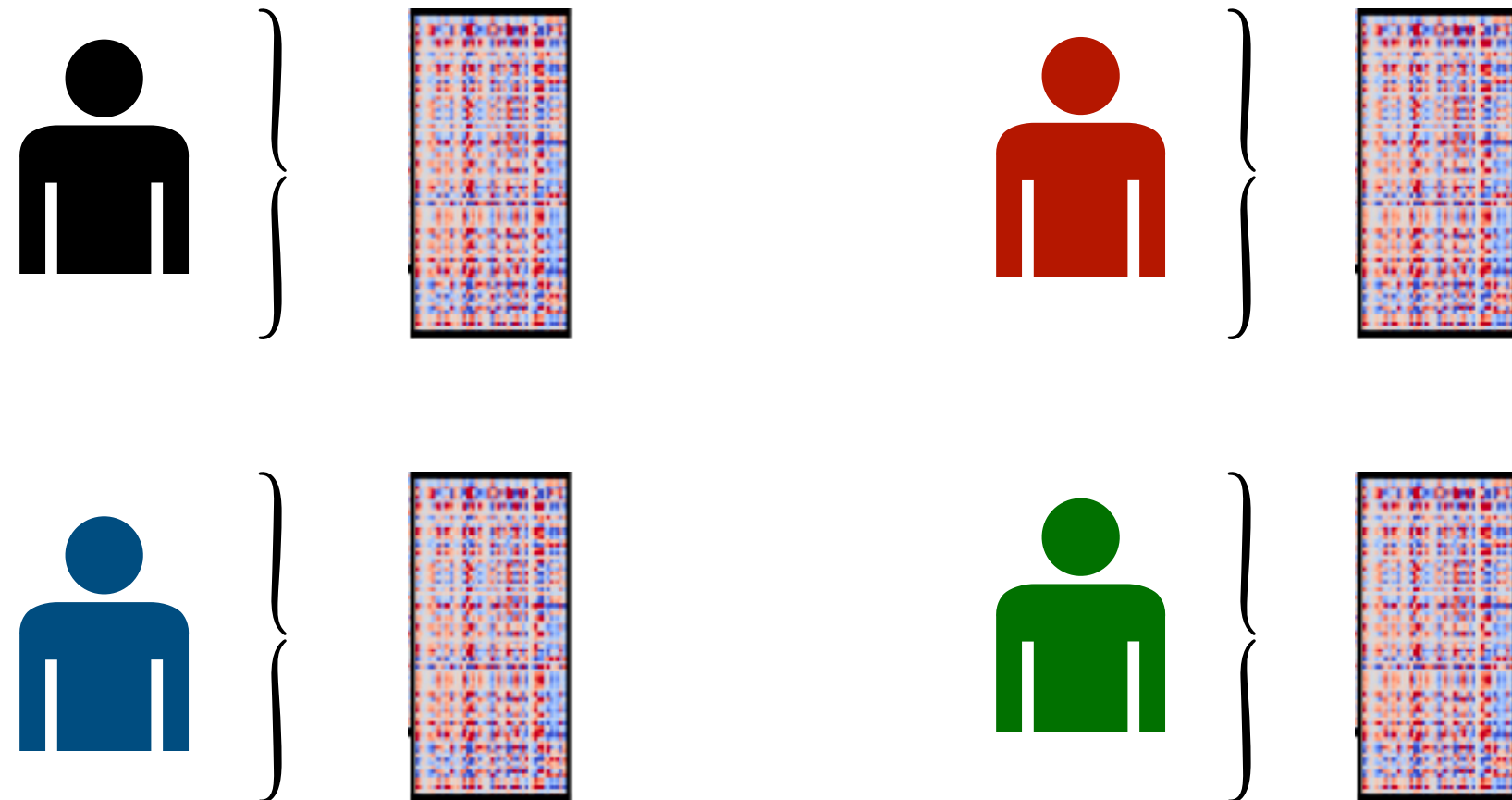
1. By representing the systems in common format through these activation matrices



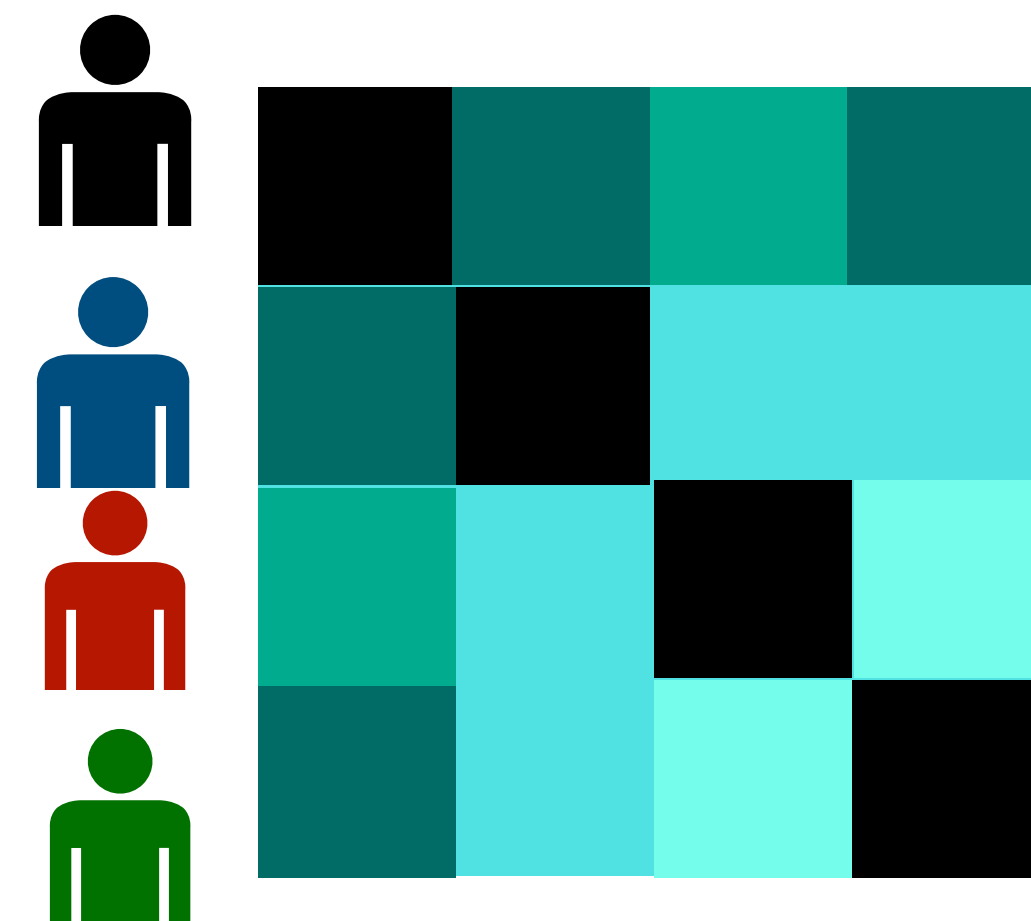
Going back to our initial question:

How to understand complex systems?

1. By representing the systems in common format through these activation matrices



2. And measuring the distance across these activation matrices



Hands on example

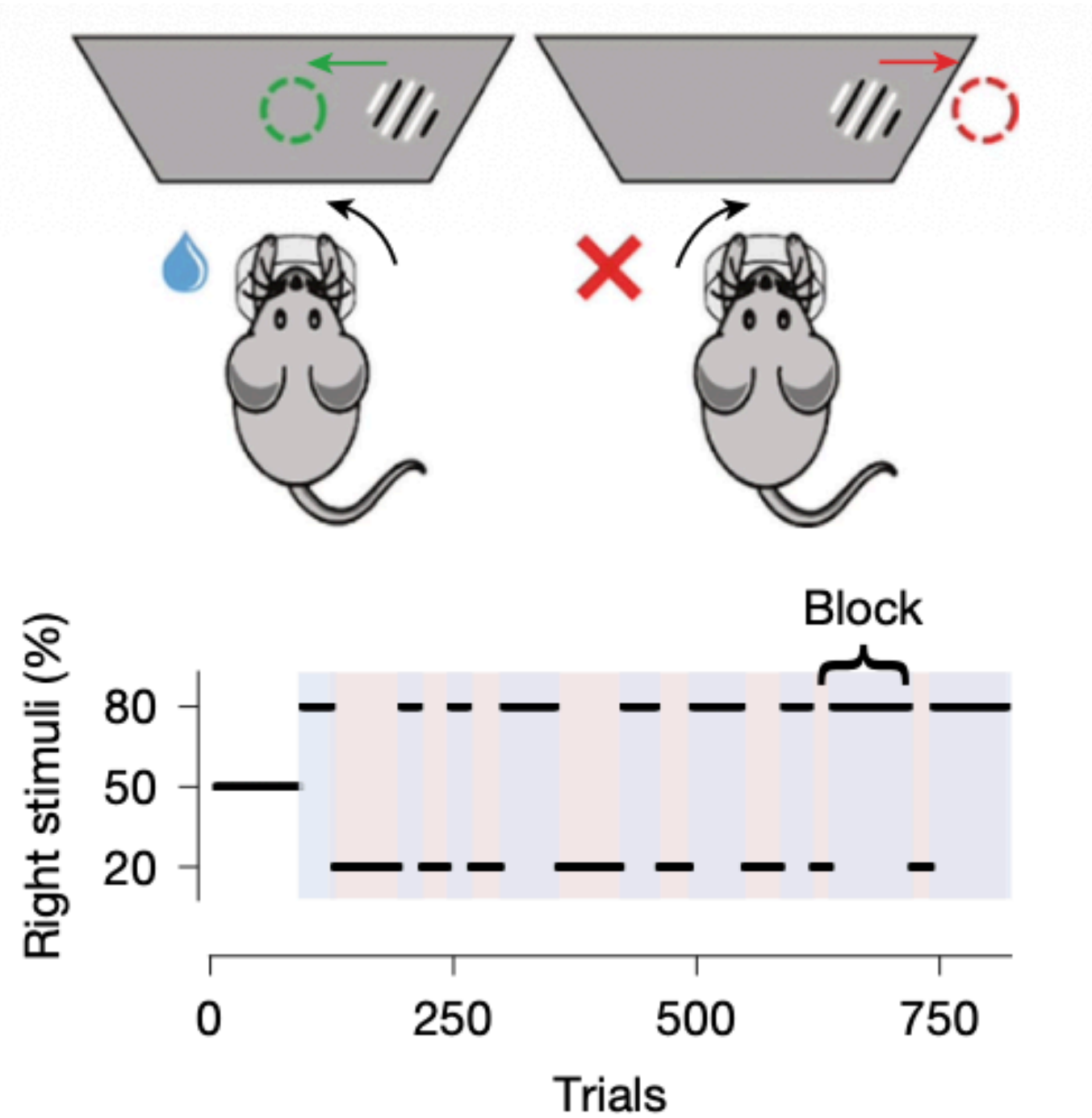
Question: Is there any structure across regions?

Dataset from IBL (Angelaki et al 2025)

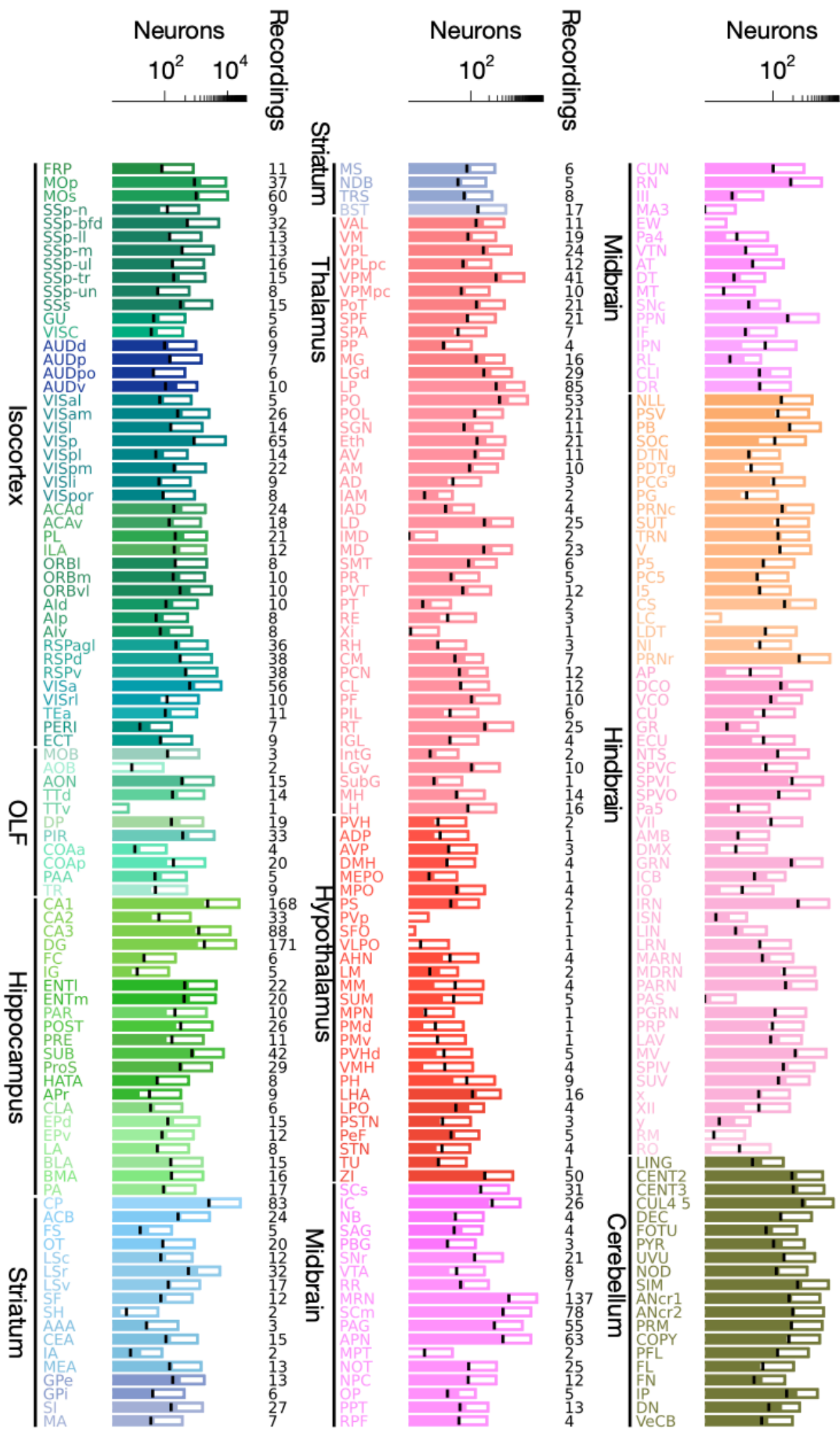
International Brain Laboratory Brainwide Map dataset (Angelaki et al. 2025)

Decision-making task:

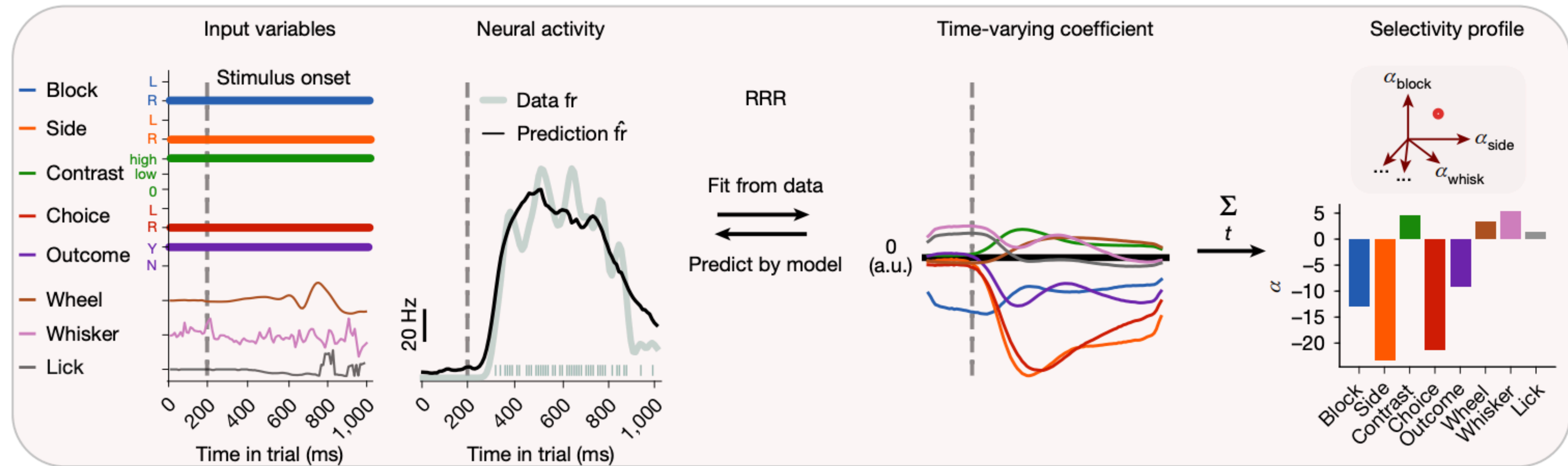
- Turn the wheel **left** or **right** depending on where the visual grating appears
- Stimulus contrast: from 100% to 0%
- Bias: probability of stimuli appearing to the right



621,733 neurons recorded with 699 Neuropixels probes across 139 mice



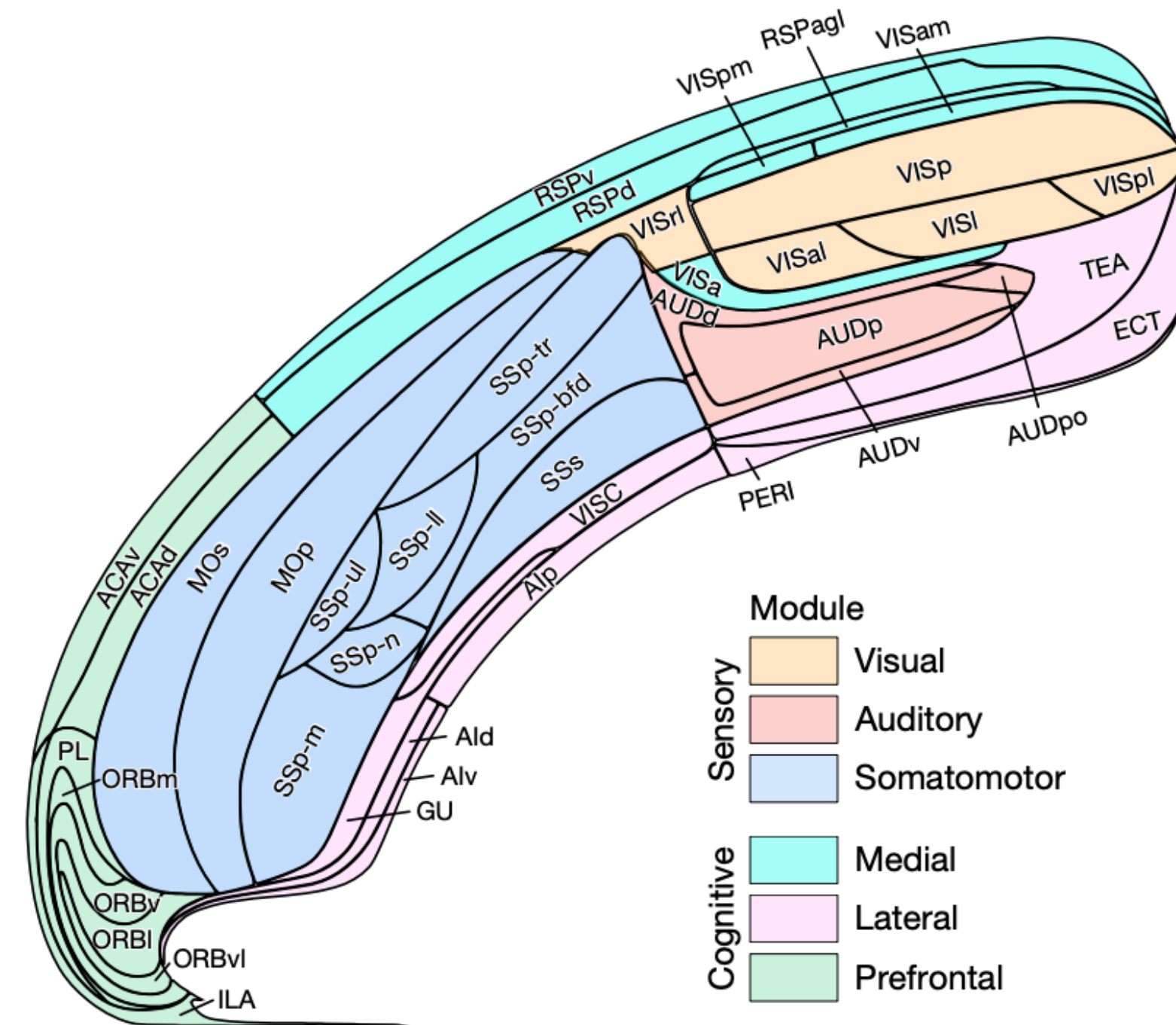
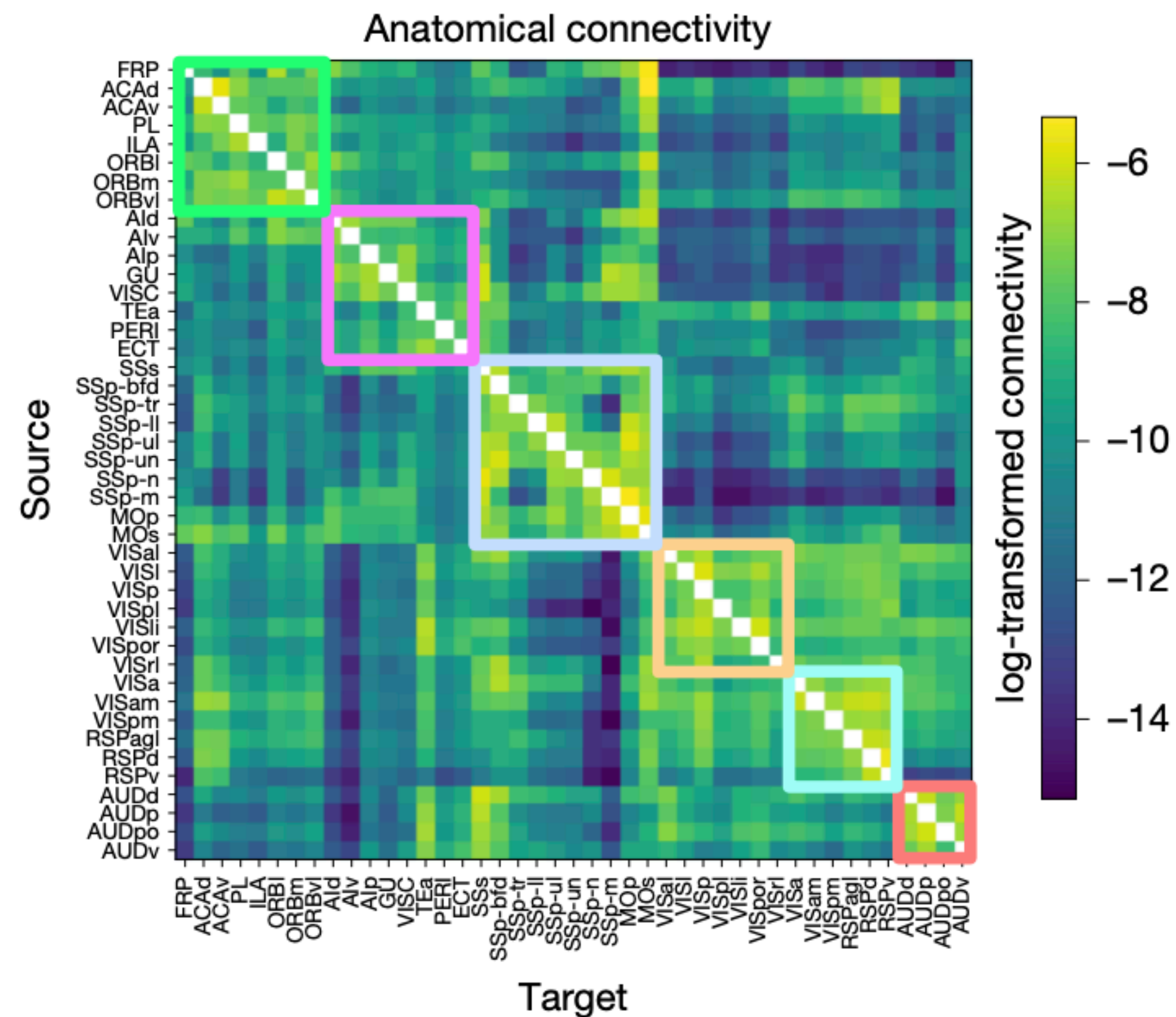
Encoding model from Posani et al (2026)



The conditions can be any task variable that might modulate the neural response

Therefore, we will be using the **predicted parameters** of an encoding model fitted on each neuron

Allen Institute Mouse Brain Connectivity Atlas



Position in the cortical hierarchy

↓

VISp
AUDp
SSp-II
AUDd
SSp-n
SSp-ul
Alp
SSp-m
SSp-un
SSp-bfd
VISl
AUDv
SSs
VISC
SSp-tr
VISli
MOP
VISrl
VISpl
RSPv
RSPd
GU
RSPagl
PERl
ECT
VISal
ILA
ORBl
Ald
VISpm
ORBm
PL
VISpor
FRP
AUDpo
TEa
VISa
VISam
MOs
ORBvl
ACAv
ACAd
Alv

Idea: find correlation between neural distances and anatomical connections

<https://github.com/brunaaf1/ShapeMetrics-tutorial/tree/main>

